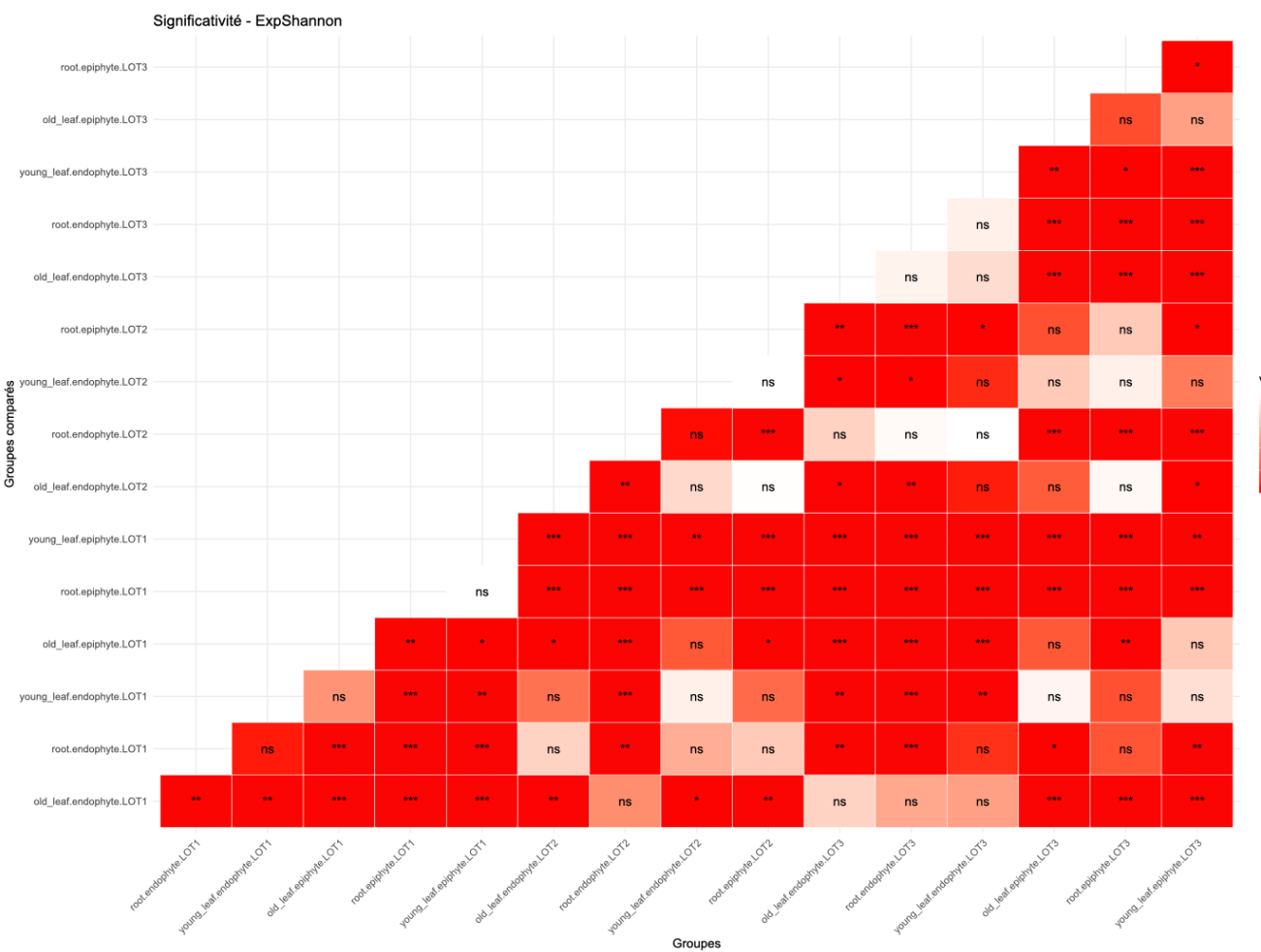
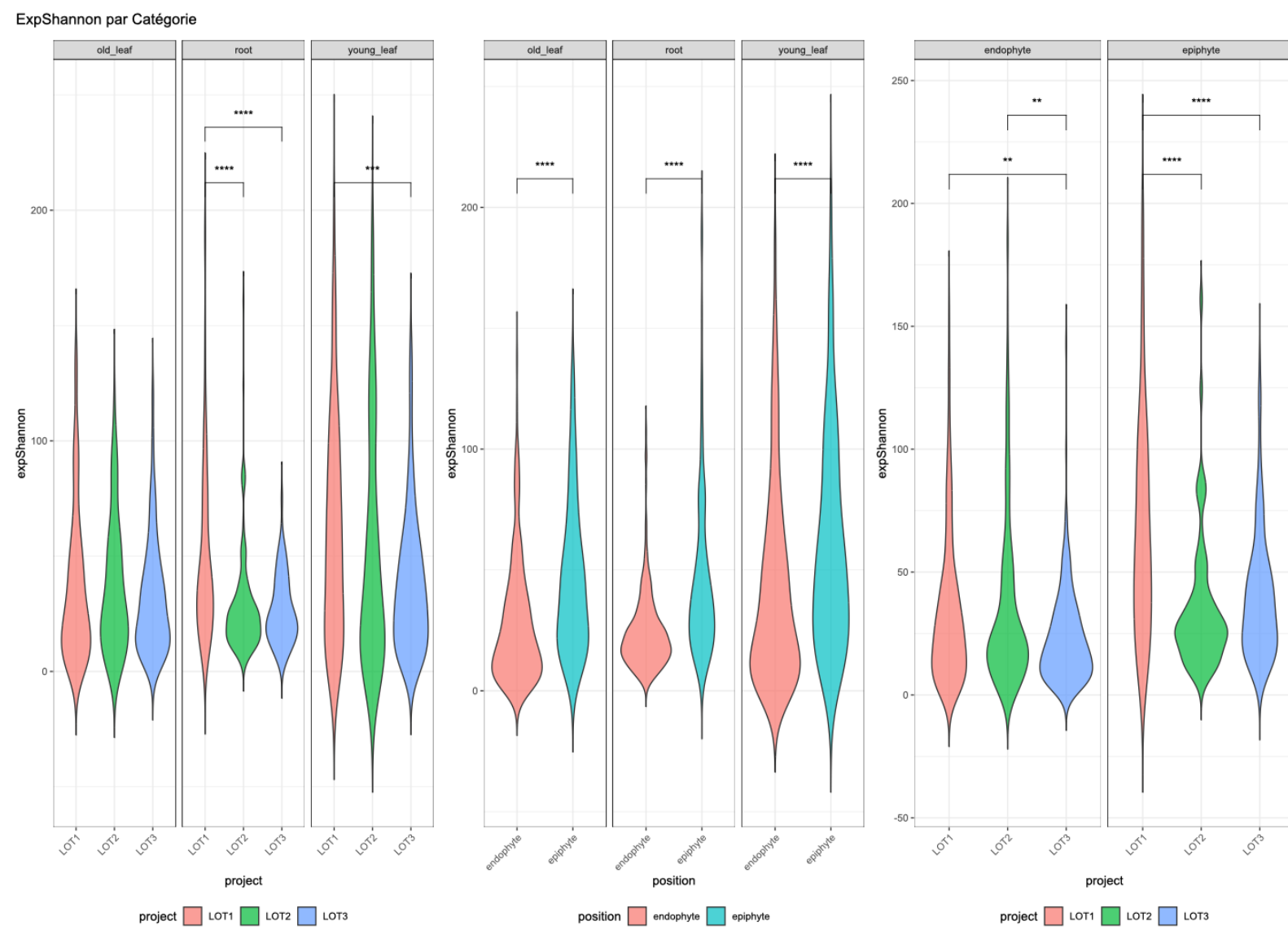
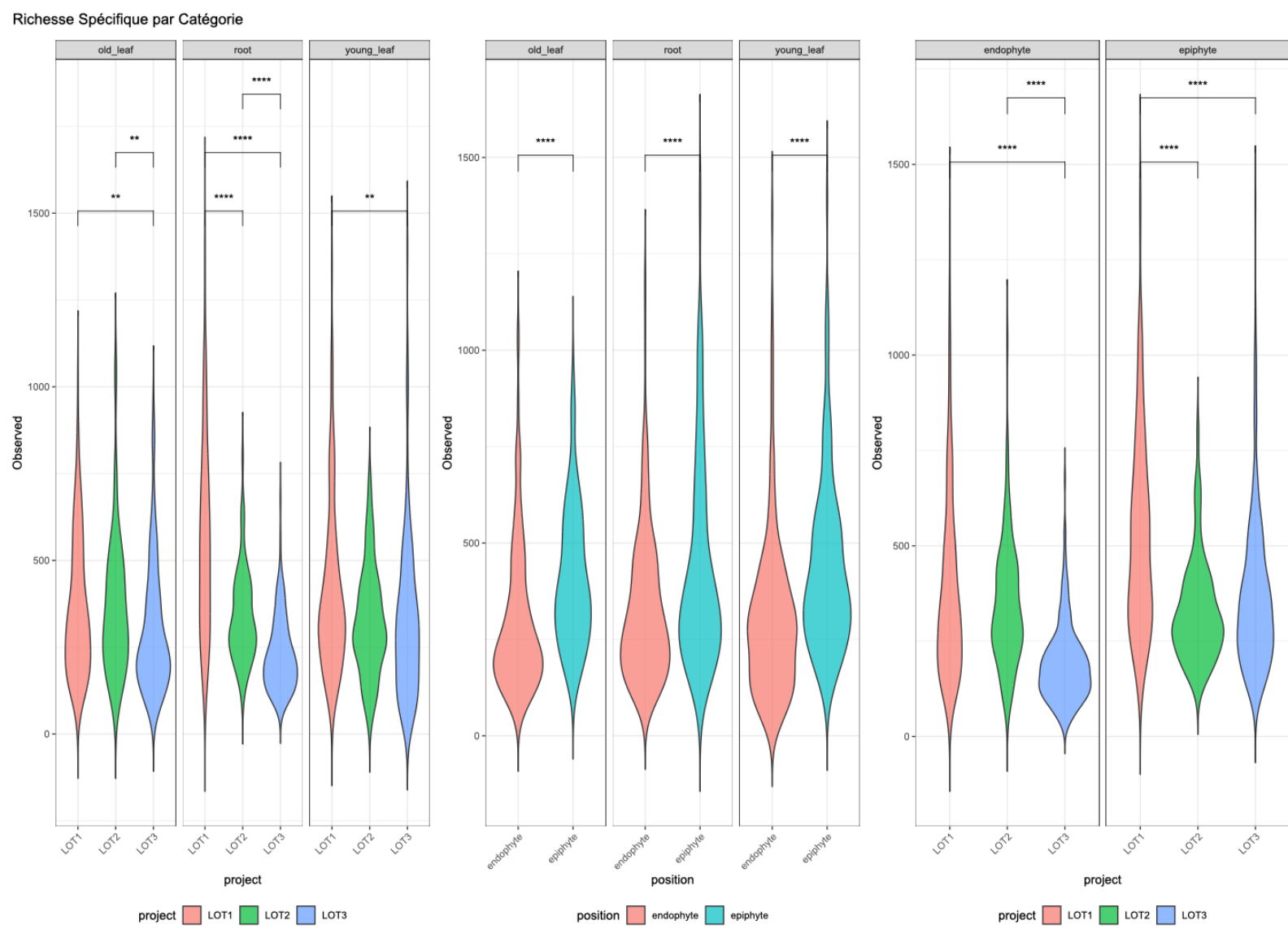
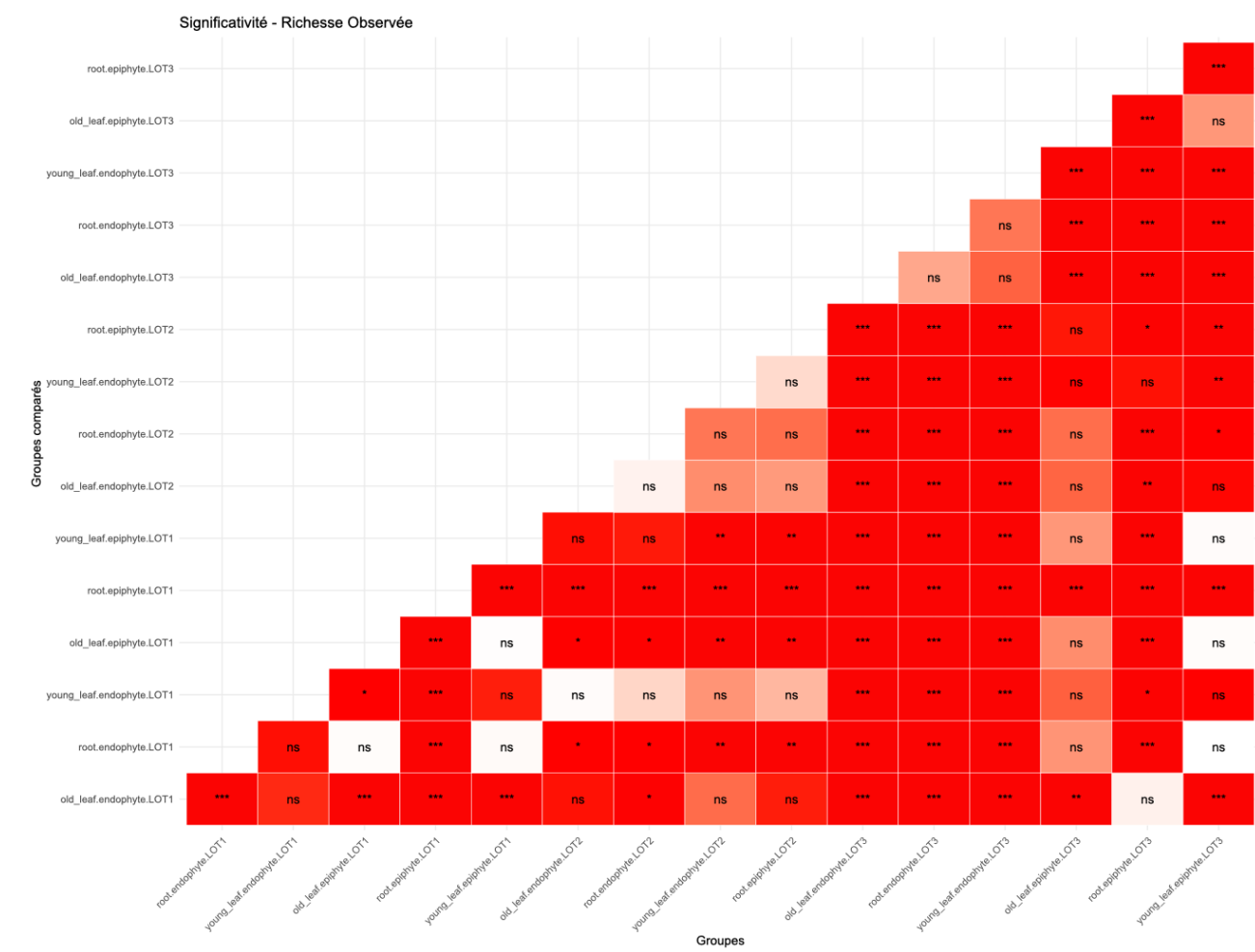


Résultats d'analyses

Analyses de diversité alpha

Plot et significativité de q=0 et q=1

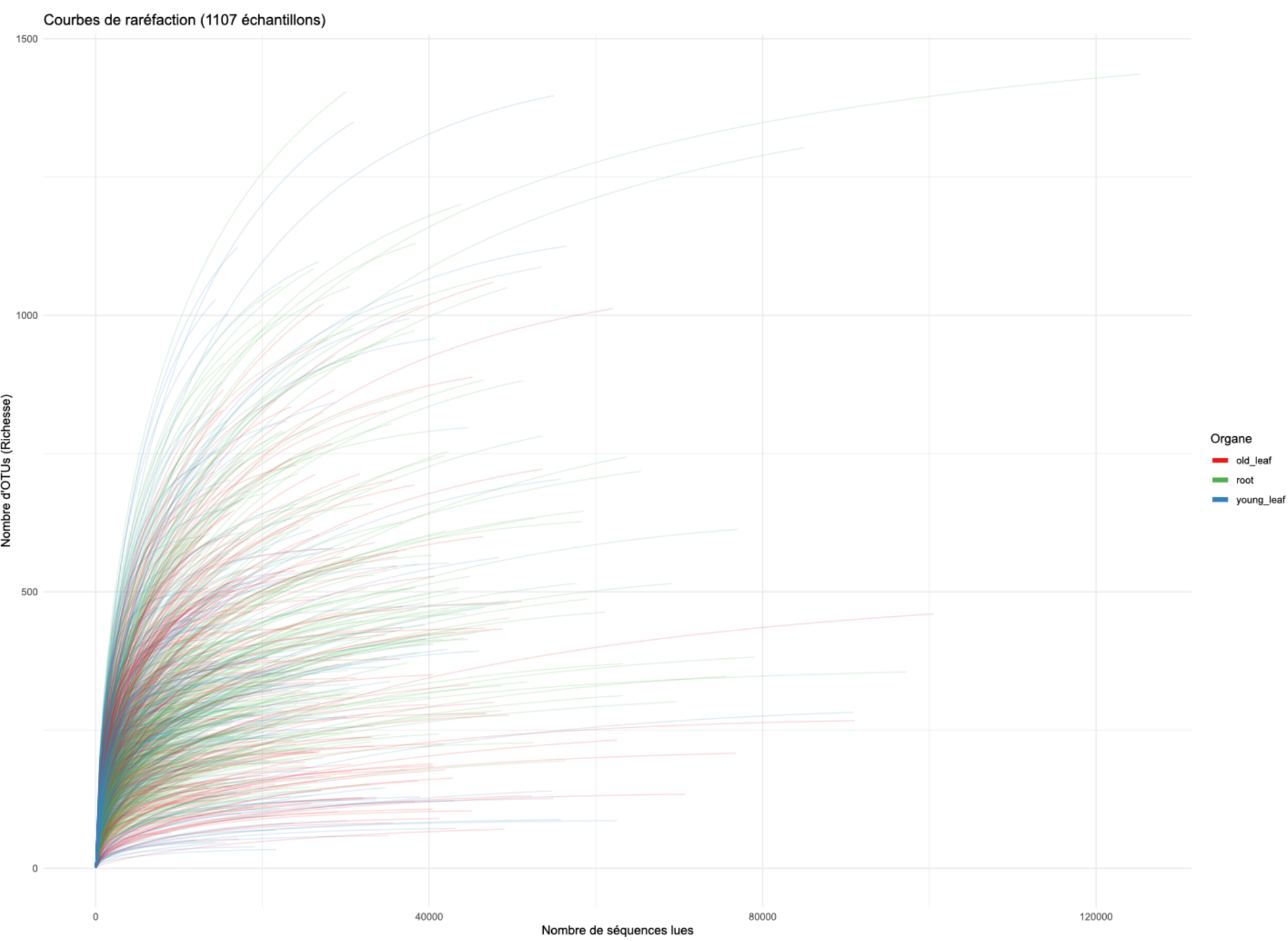


La rs globalement plus élevée dans vieilles feuilles, en epiphyte et dans le LOT1.

Diversité plus élevée dans les jeunes feuilles epiphytes du LOT 1.

A bien détailler mais pas forcément très perspicace...

Vérification quantitative de l'échantillonnage



Globalement les phases de plateau sont atteintes (ou presque) donc l'échantillonnage et le nombre de séquences lues paraissent suffisantes pour des analyses robustes.

Je mets une réserve sur le graphique de diversité, normal de voir des oscillations pour une courbe ?

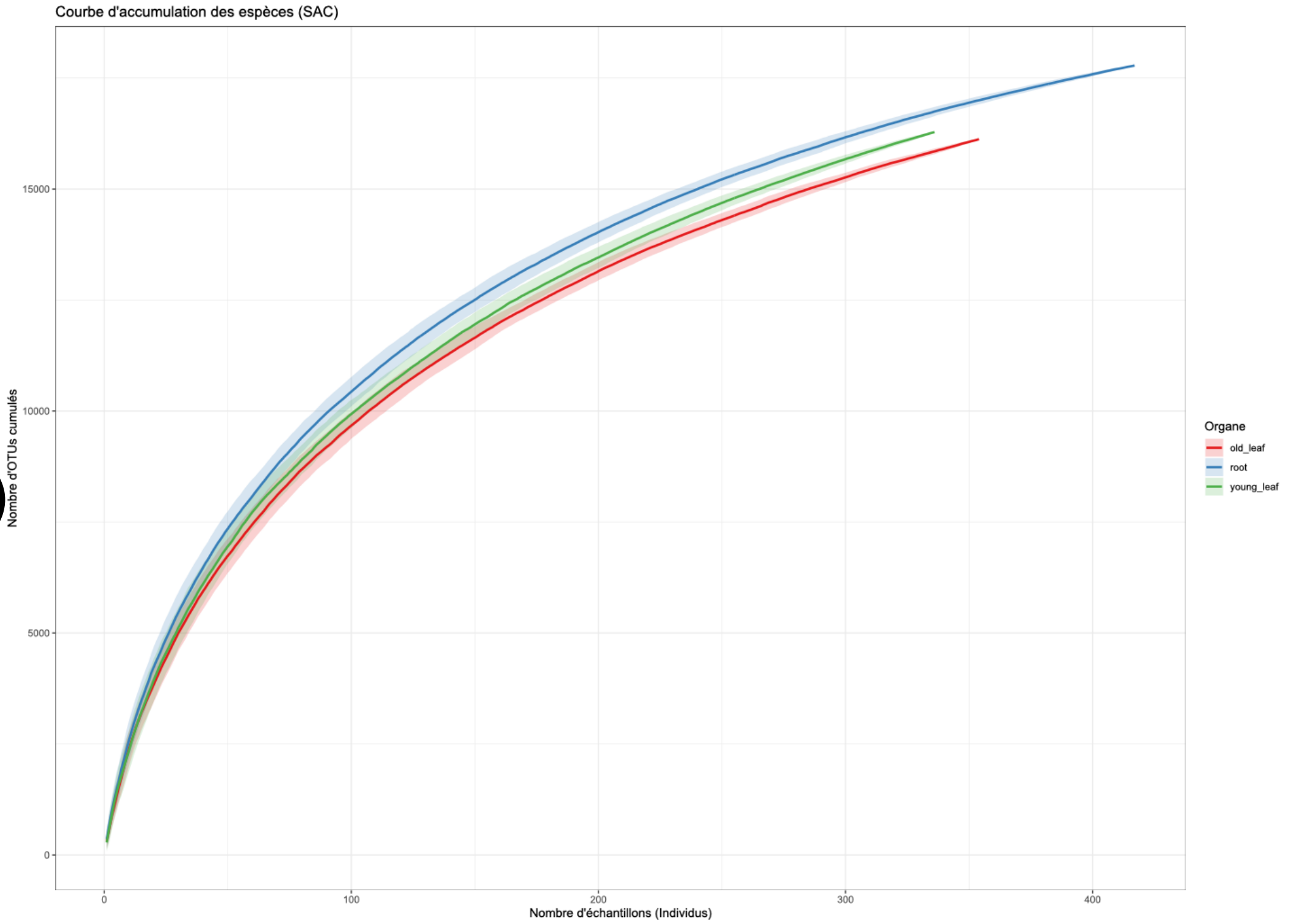
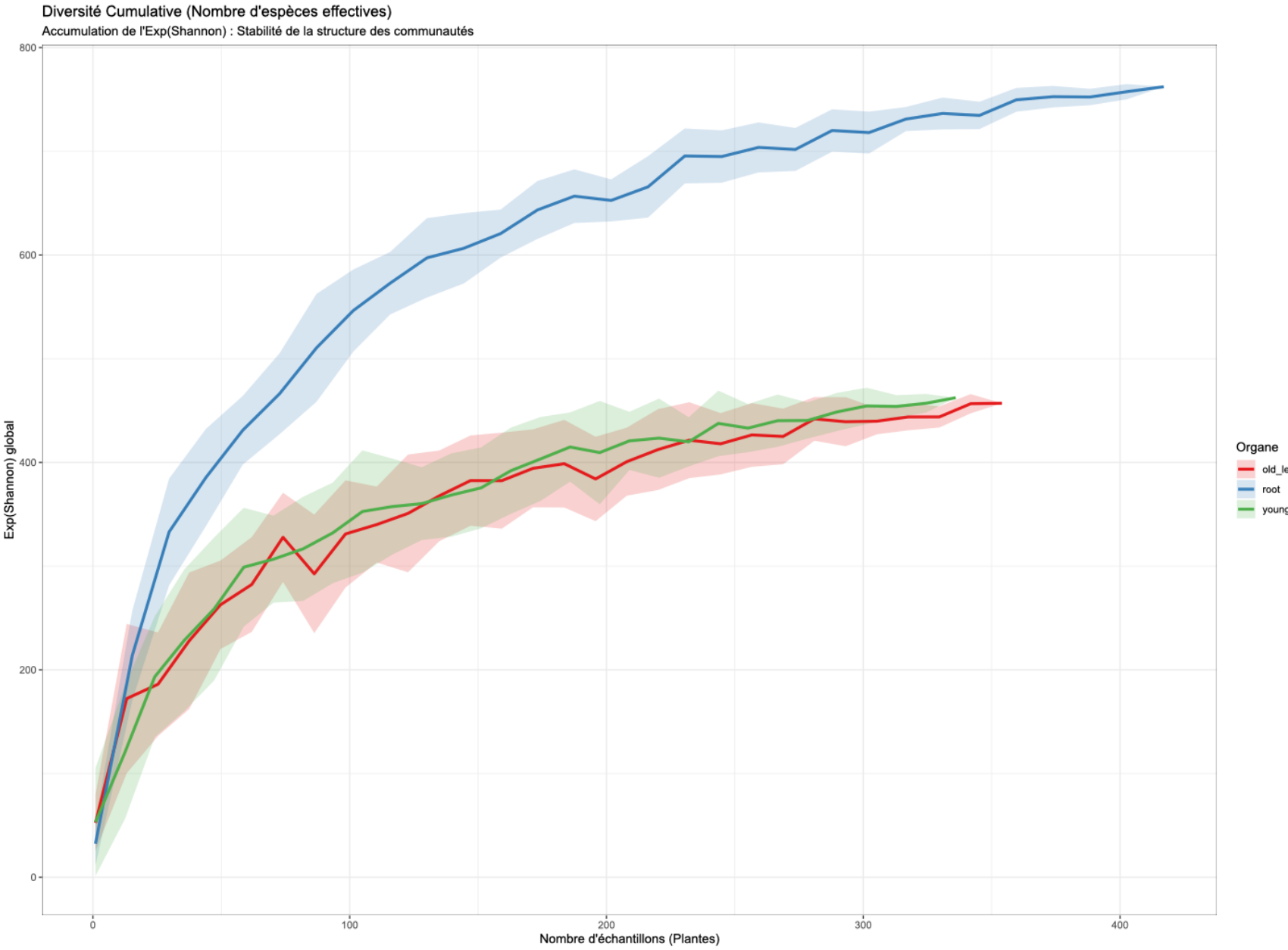
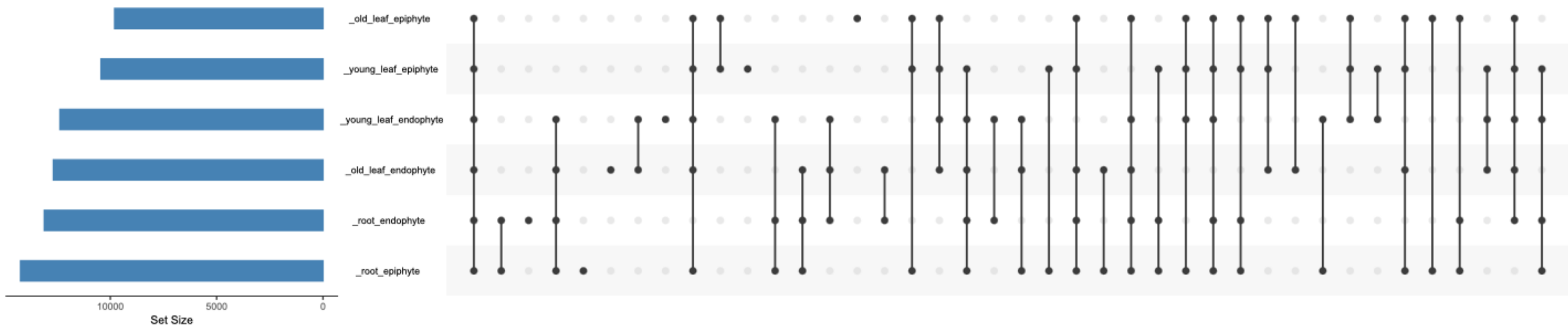
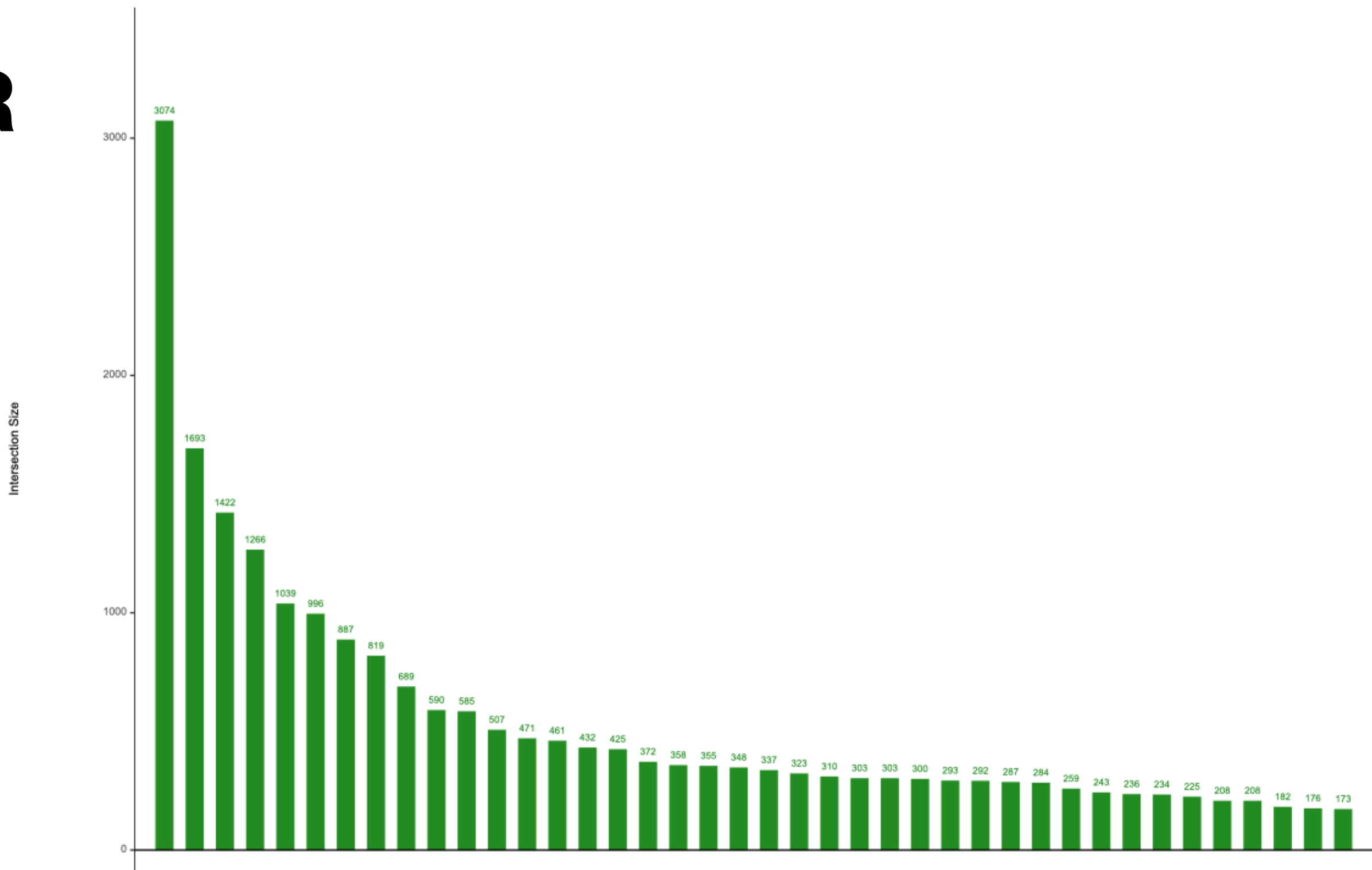
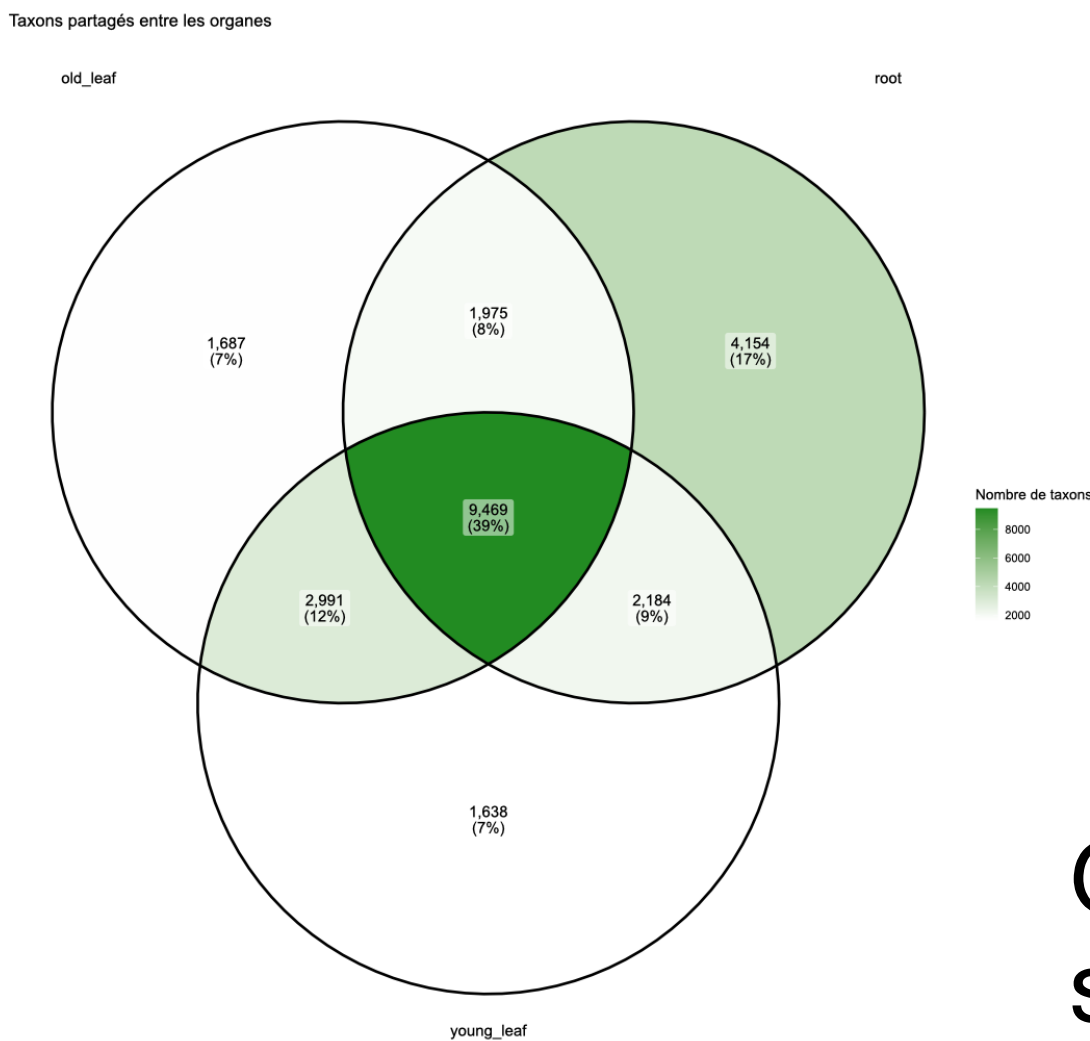
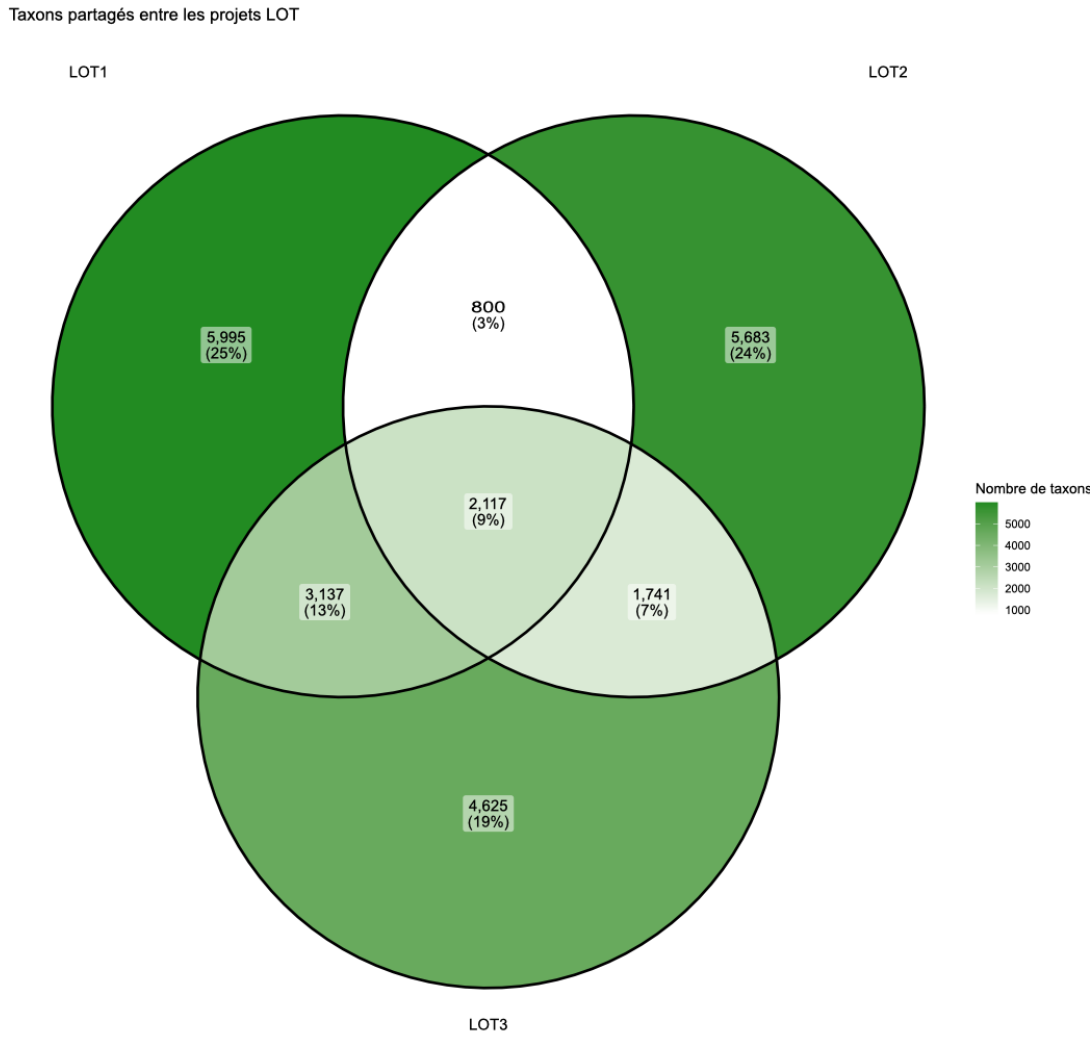
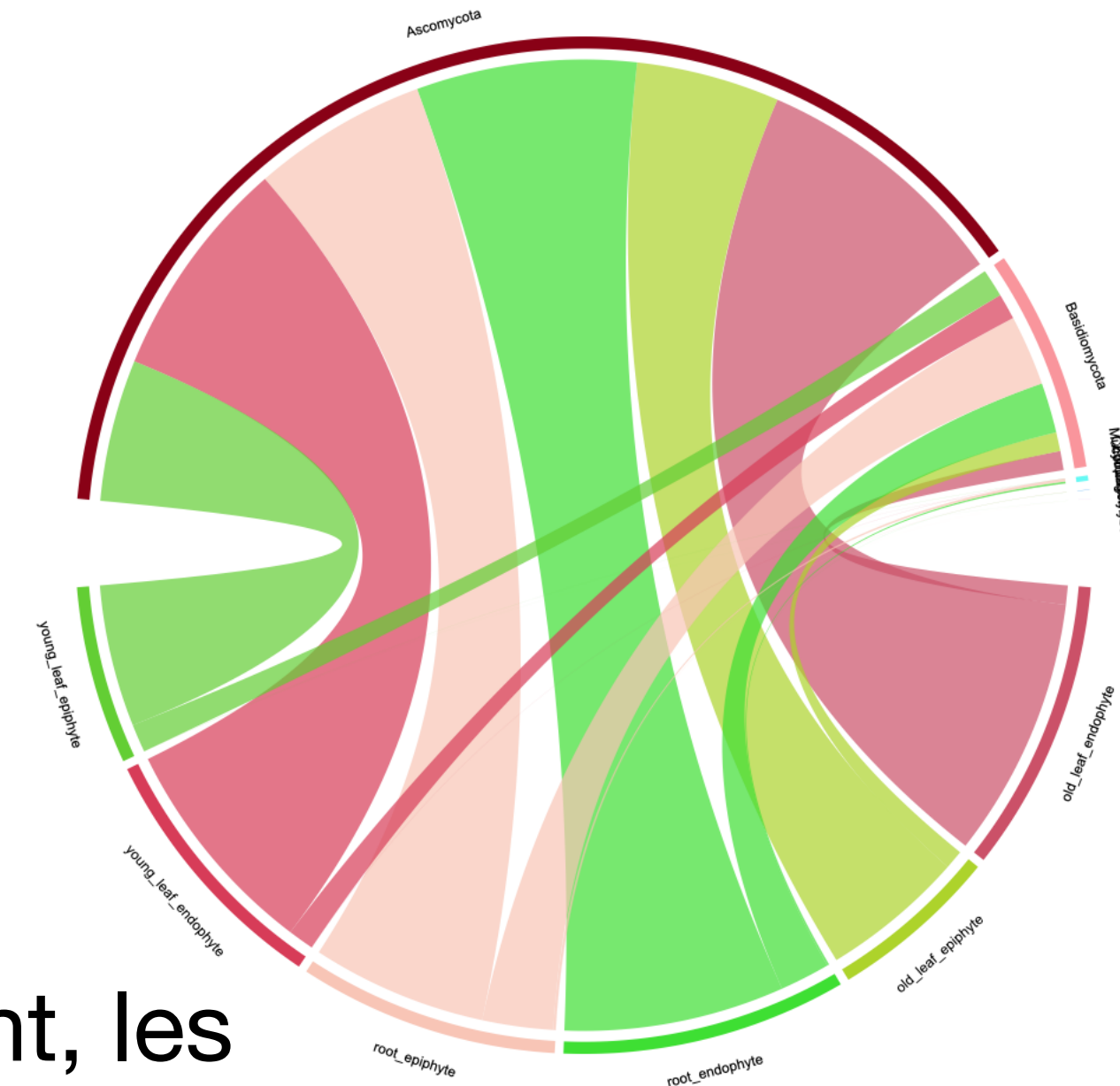
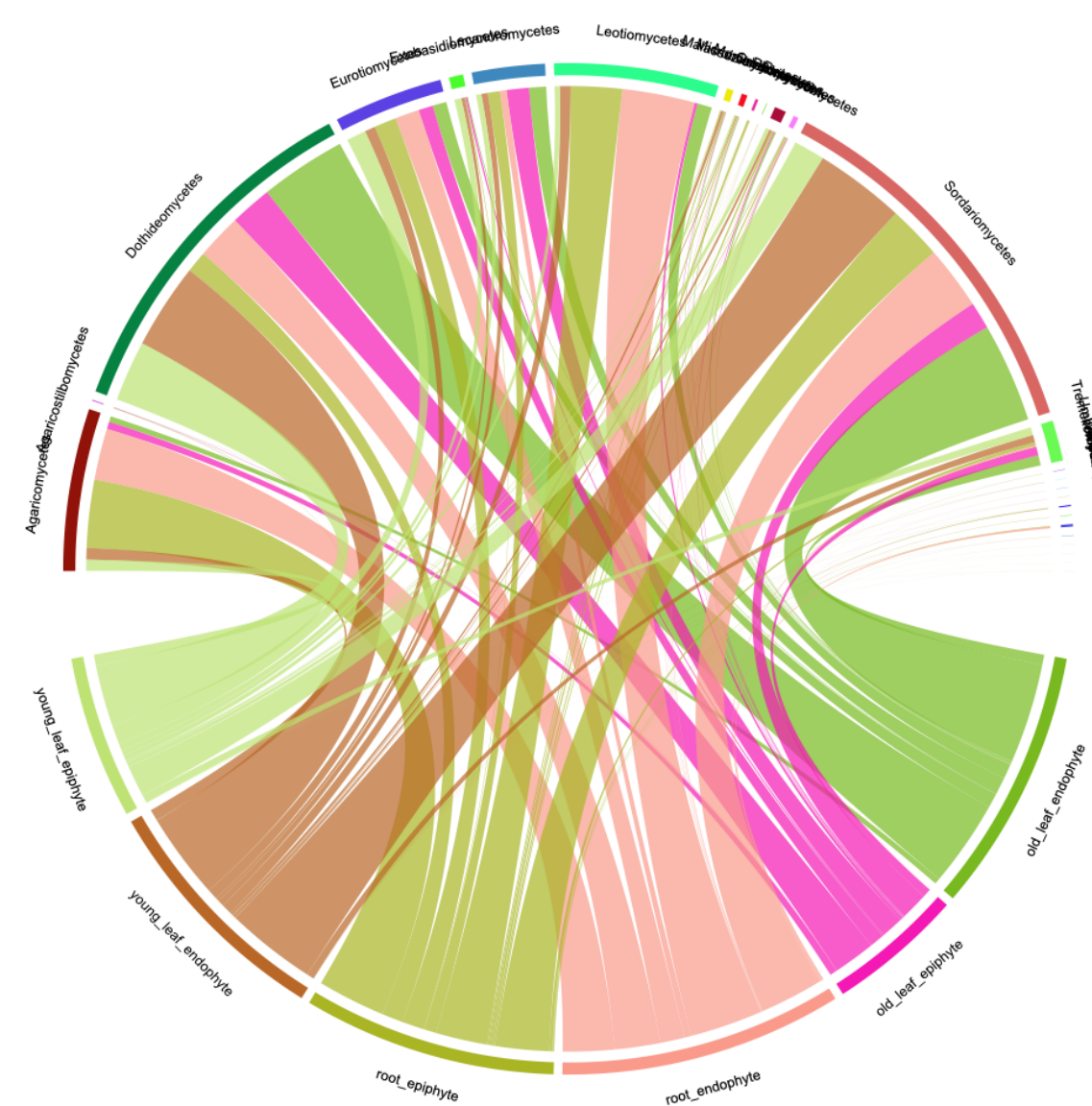
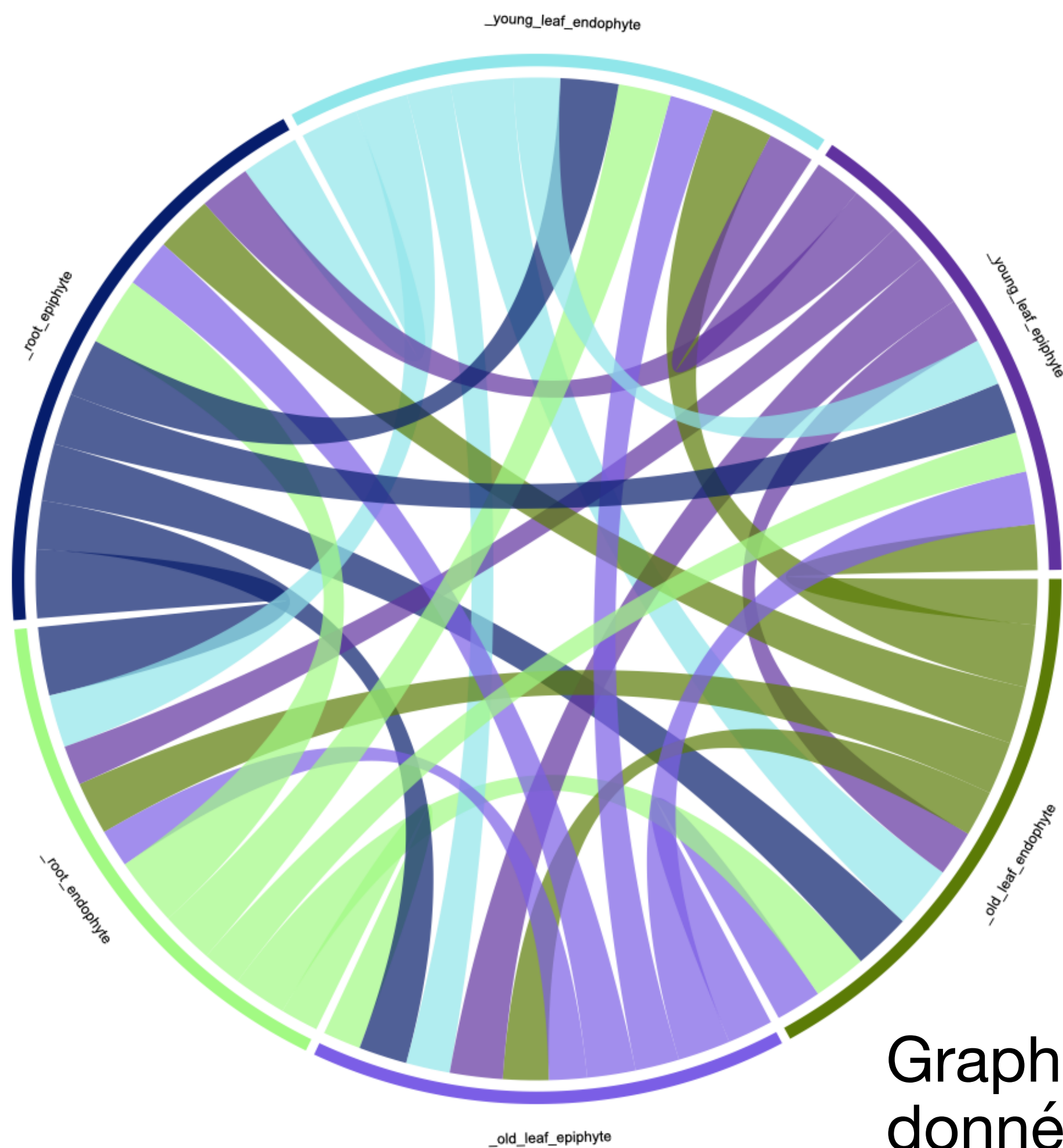


Diagramme de Venn et UpSetR



Graphiques assez simple a décrire, ils permettent d'avancer des hypothèses sur le core microbiota. Bcp d'otu spécifiques au milieu (projet), majorité d'otu partagés entre organes

Diagrammes de Chord



Graphiques pas forcément pertinents étant donné les hypothèses posées précédemment, les données se retrouvent presque toutes dans les mêmes organes et les éventuels spécialistes rares sont masqués

Analyses de diversité Beta

Ordinations (NMDS, k=3, stress=0.13)

Stress:

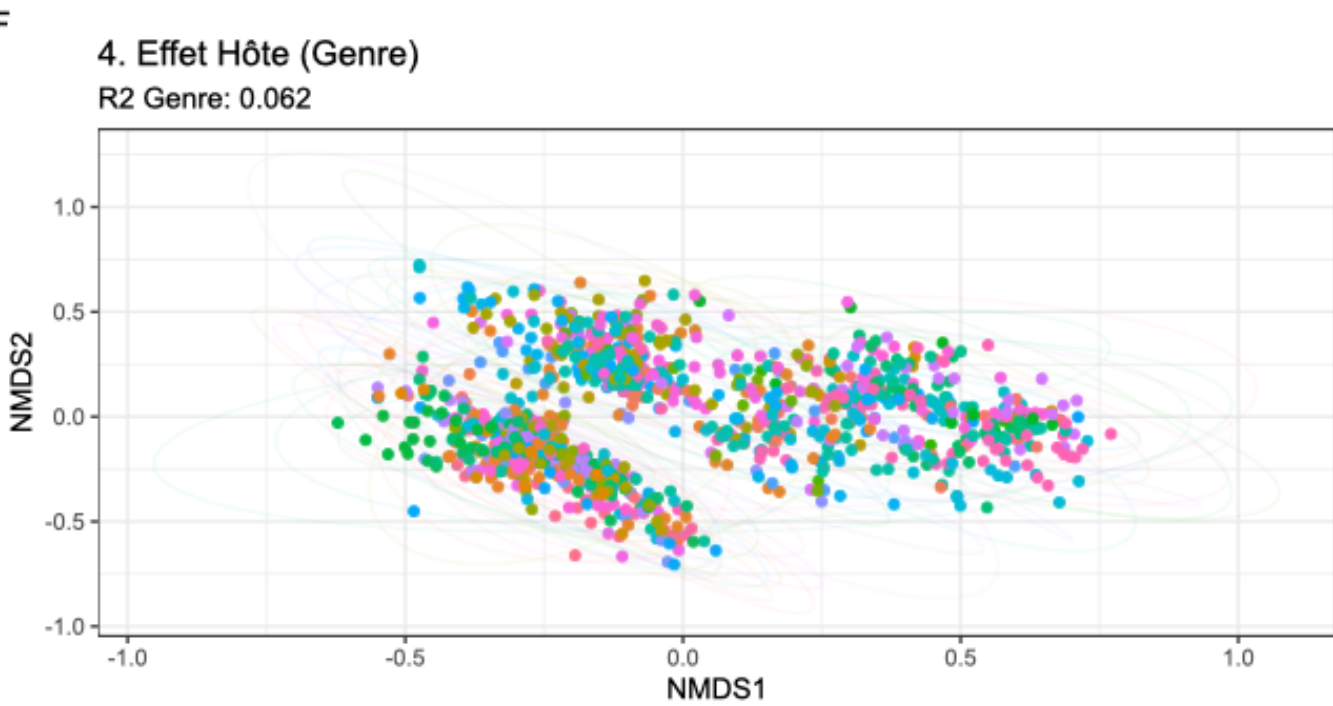
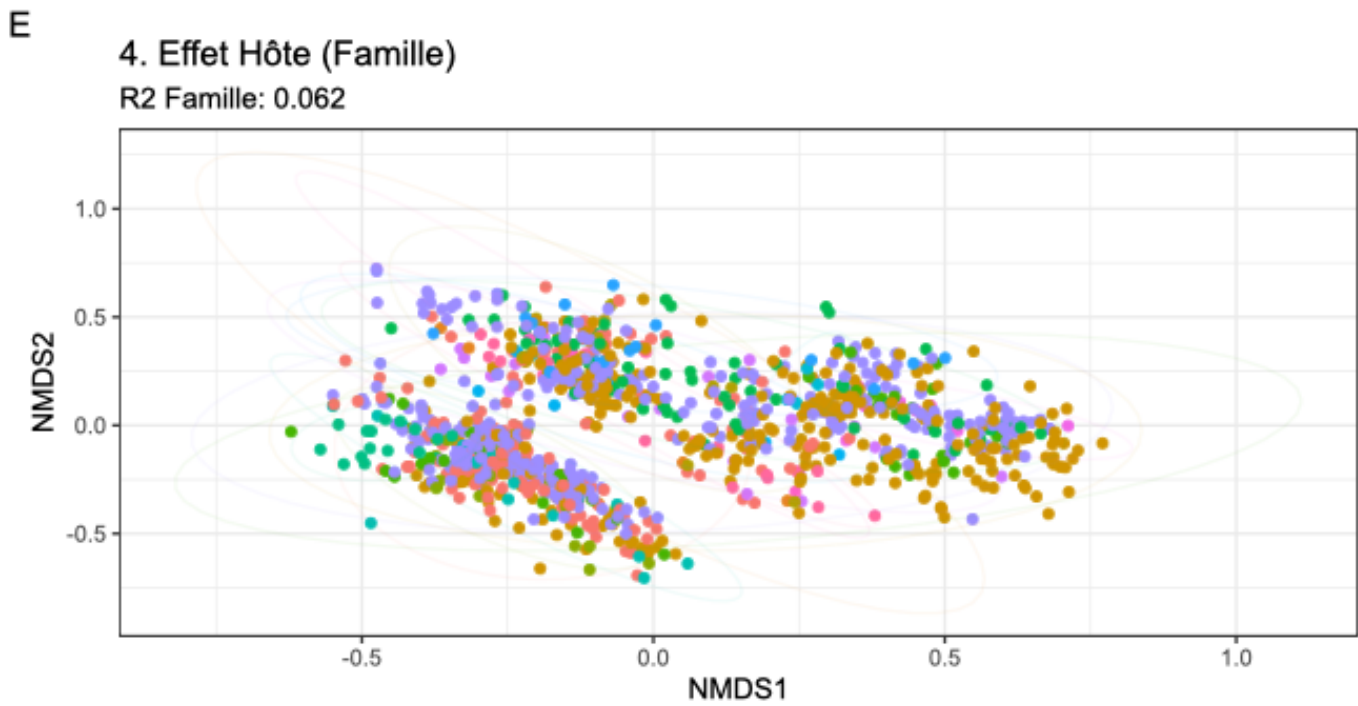
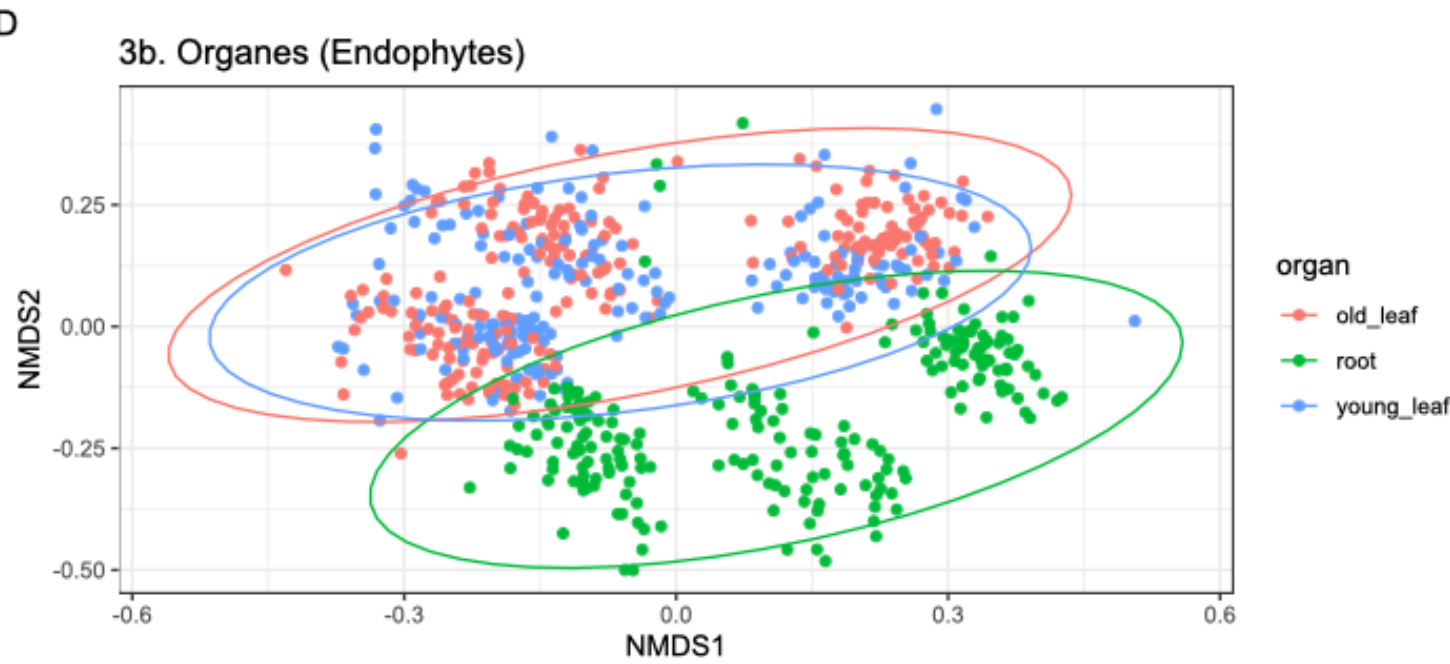
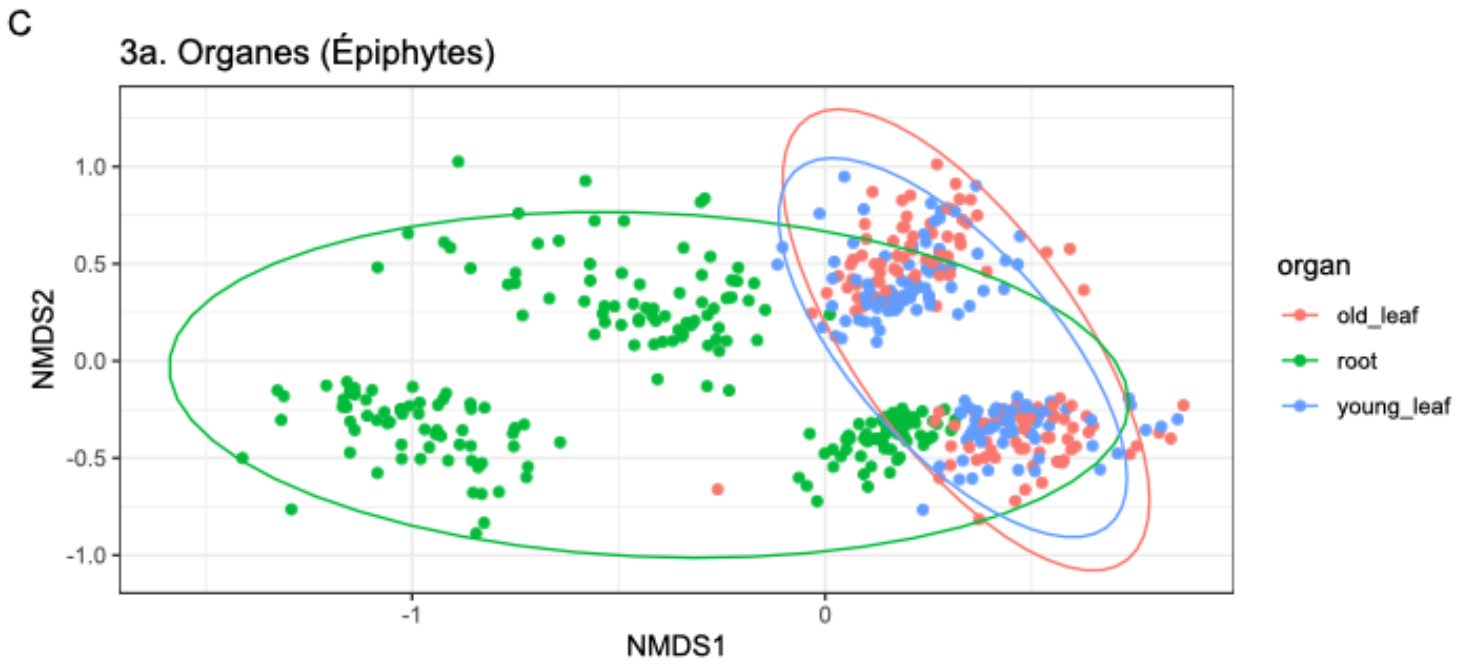
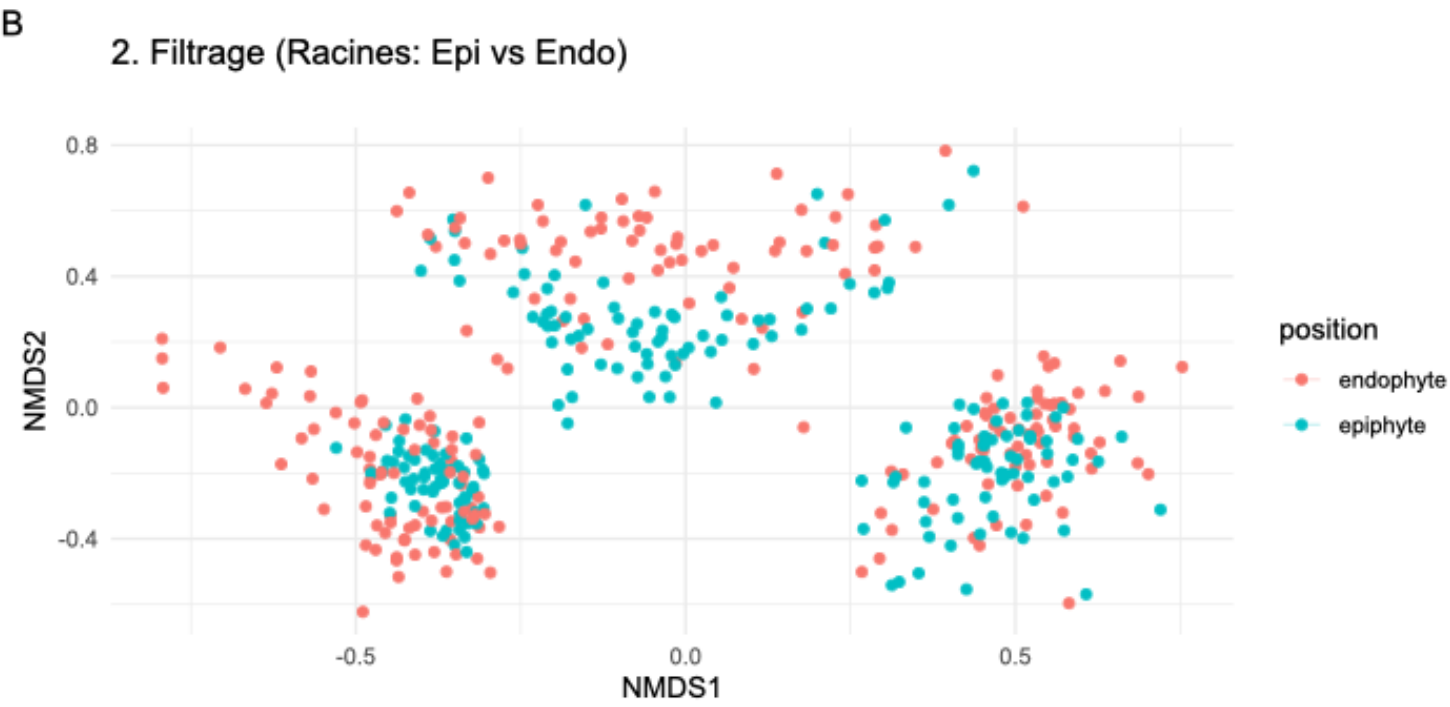
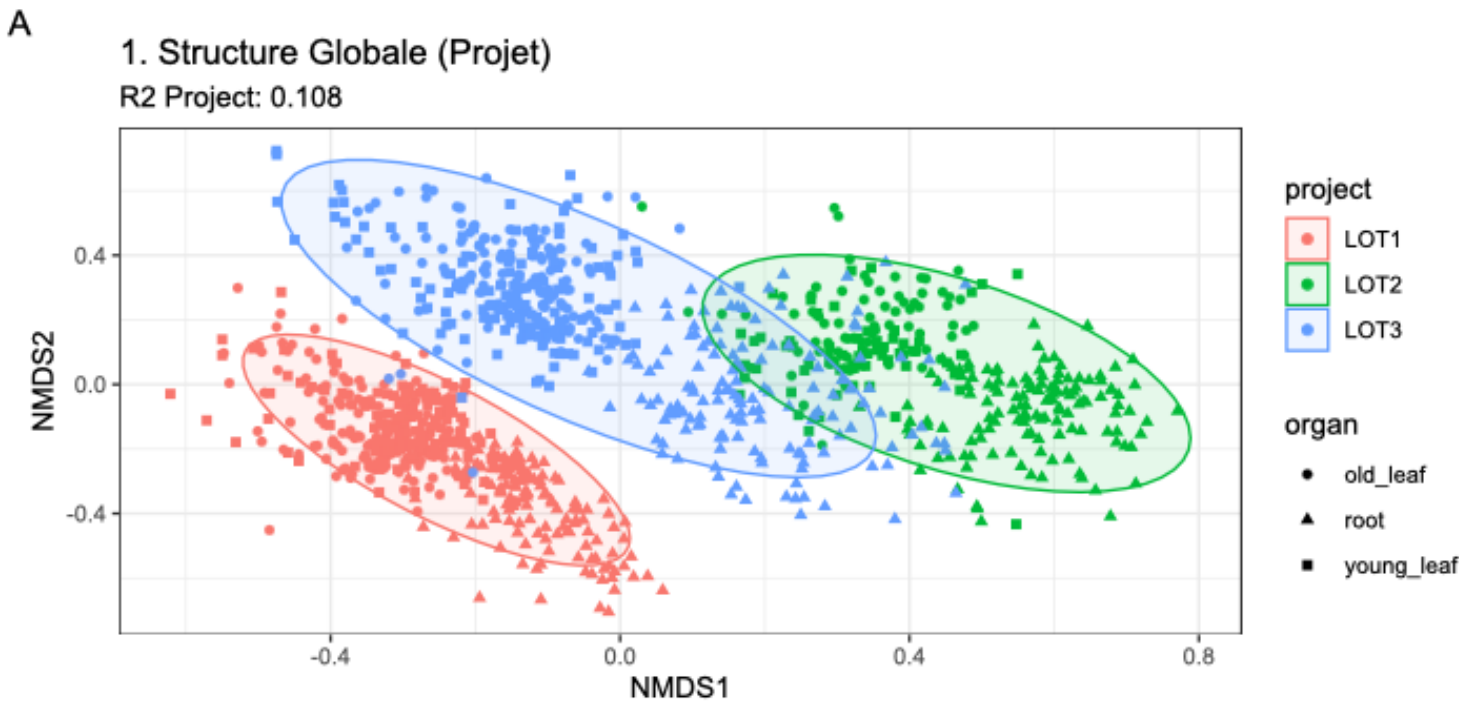
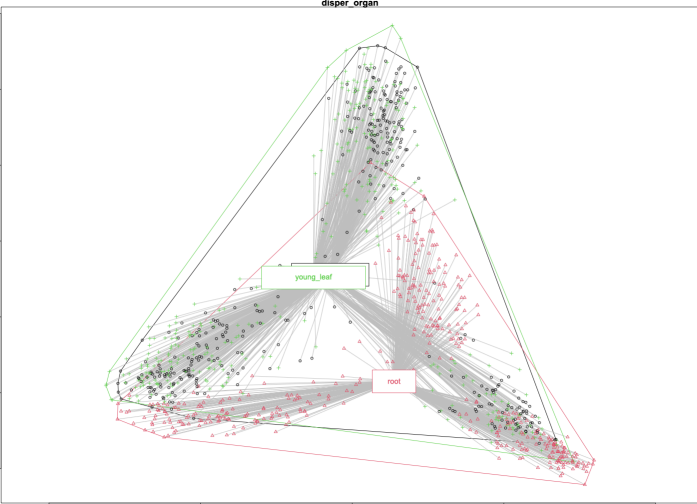
- Global : 0.1363989, k=3
- B: 0.1783157, k=2
- C: 0.1592424, k=2
- D: 0.1592424, k=2
- E: 0.2286184, k=2
- F: 0.2286184, k=2

Permanova

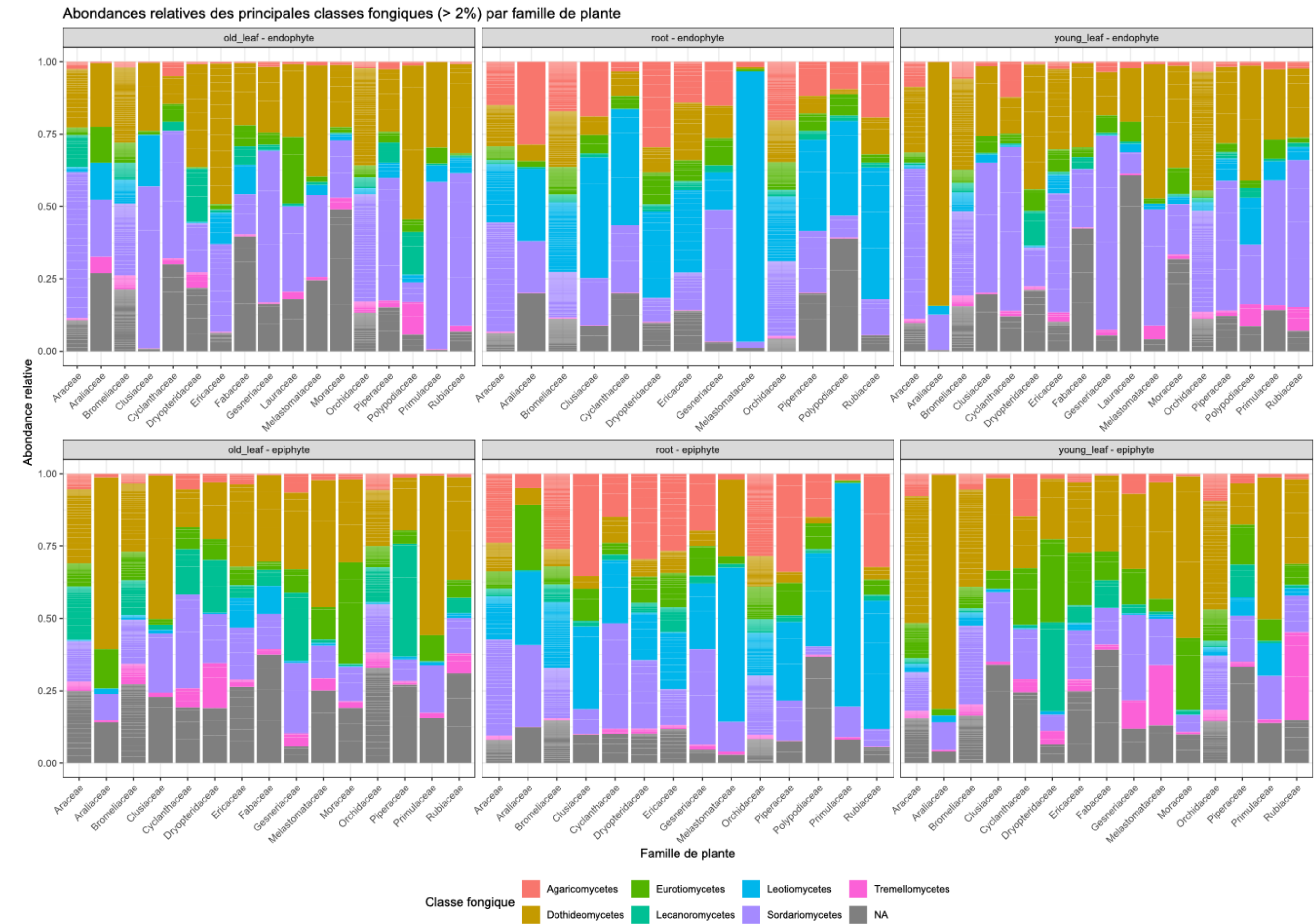
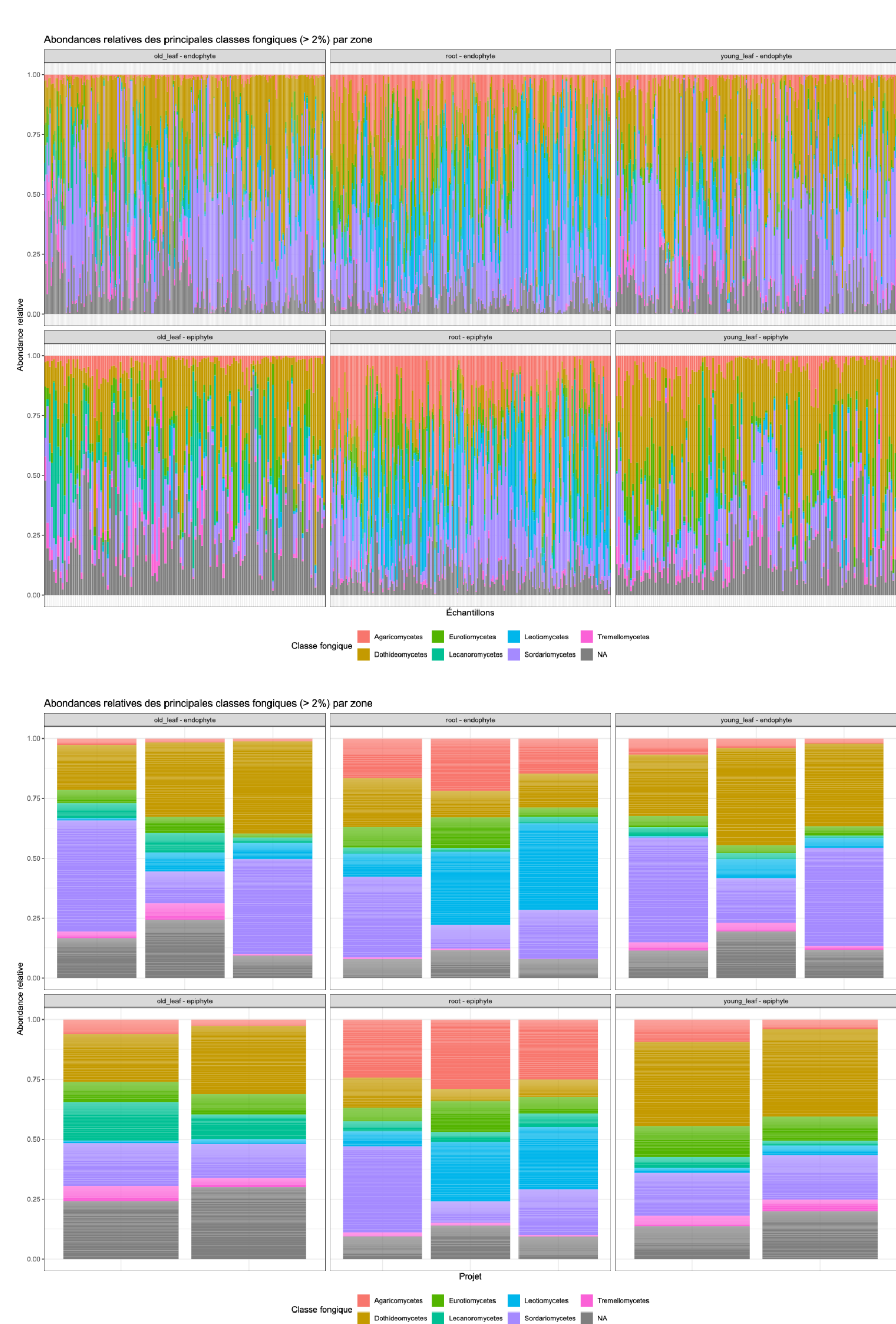
	Facteur	Df	R2	p_value
1	project	2	0.1075	0.01
2	organ	2	0.0384	0.01
3	position	1	0.0134	0.01
4	plant_family	16	0.06239	>0.05

On retombe sur les memes conclusions, effet tres fort du projet, effet significatif de l’organe et de la position mais faible, pas d’effet significatif de la famille

Dispersions vérifiées

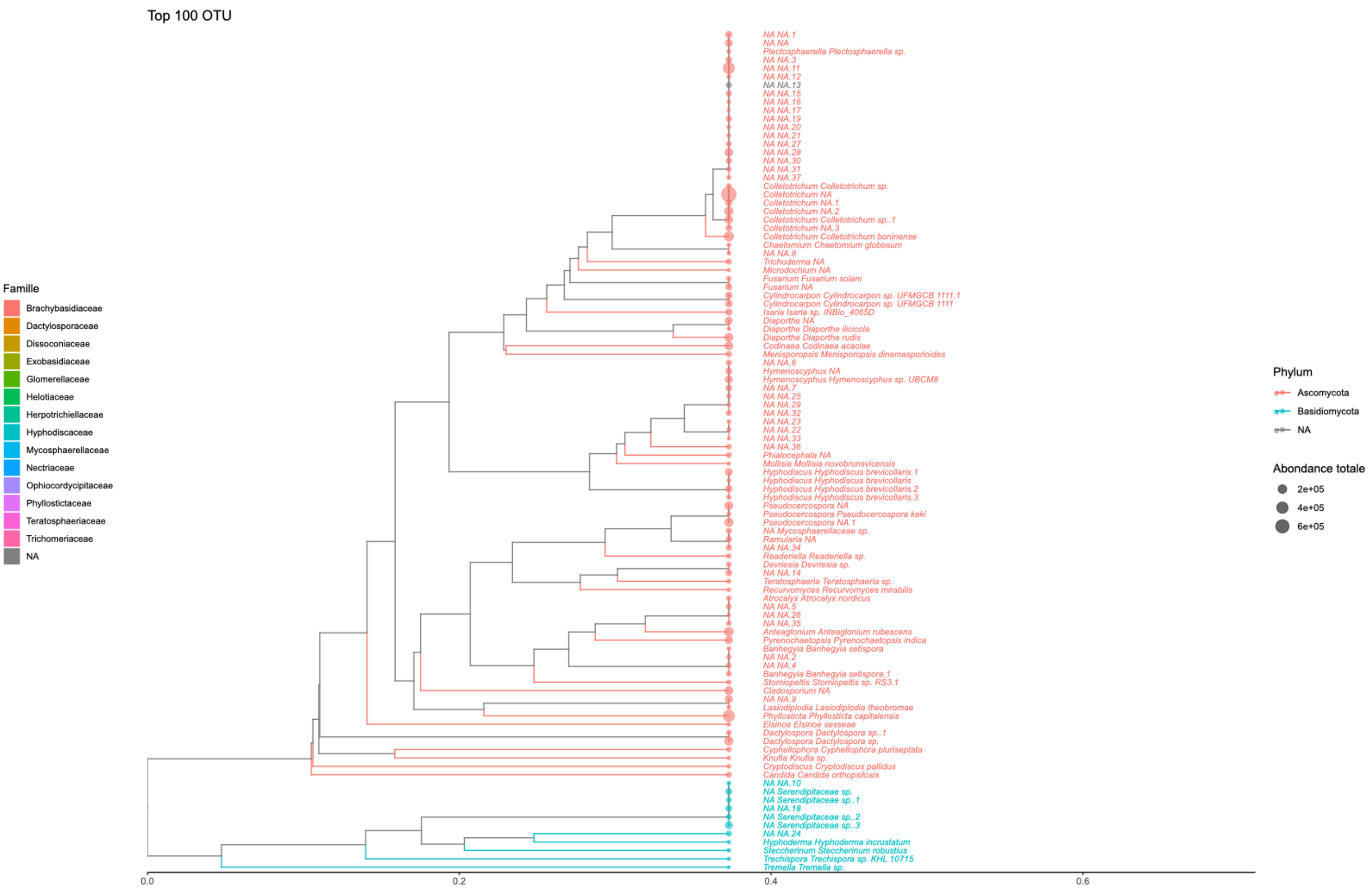
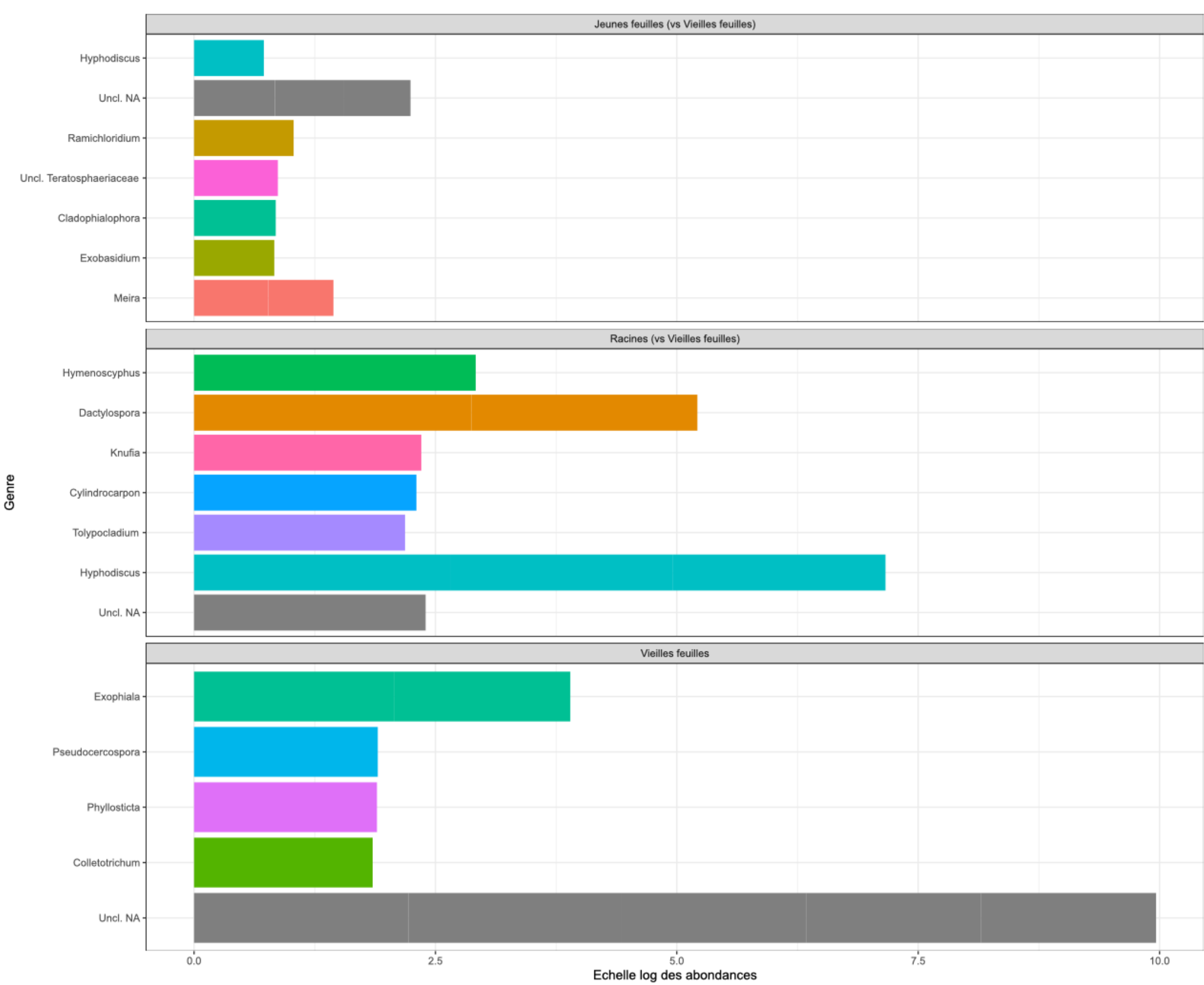


Abondances relatives des classes fongiques



Certains pattern intéressants: surreprésentation des Agricomycetes (dont le systèmes de réseau est très interessant mai en lien ?) et Leotiomycetes (pathogènes des plantes) dans les racines, avec peu de Dothideomycetes (classe très variée)

Taxons différentiellements abondants entre les organes et arbre des 100 OTU les plus abondants

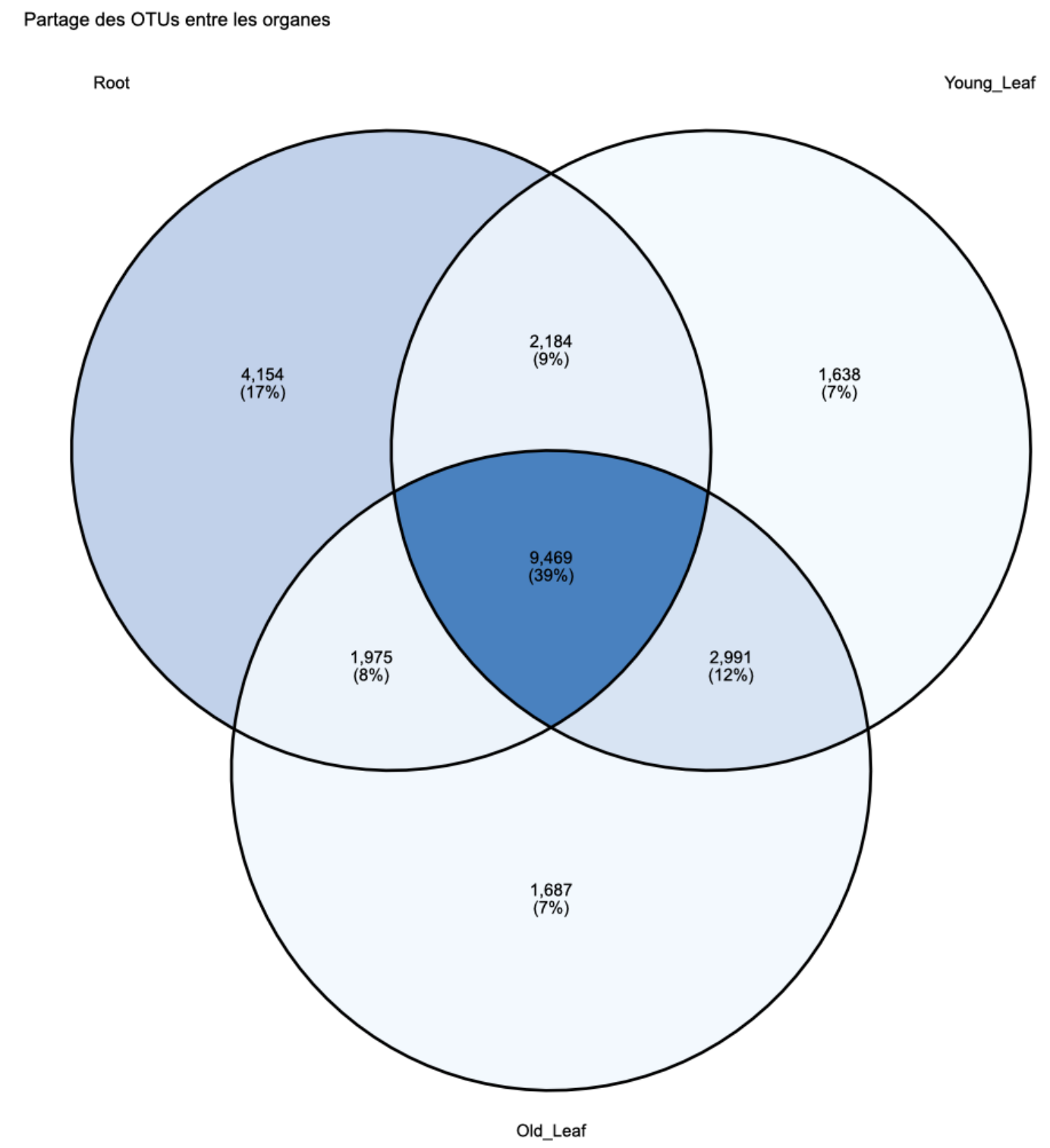


Classé uniquement avec taxonomie, séquences dispo ?

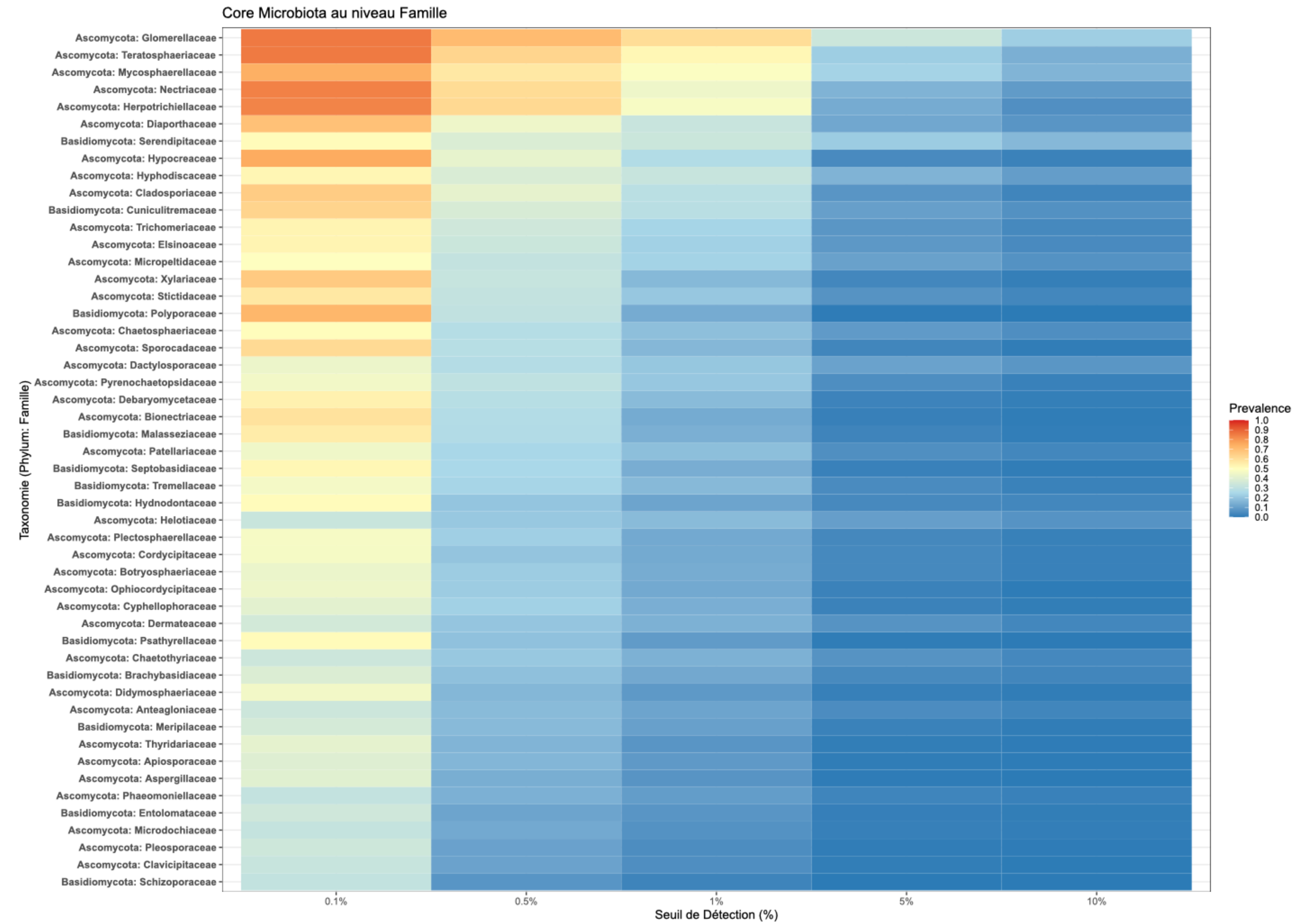
Visualisation plus explicites des genres fongiques majoritaires entre les organes

Et un arbre qui ne vaut pas grand chose mais pourrait être utilisé ultérieurement....

Core microbiota

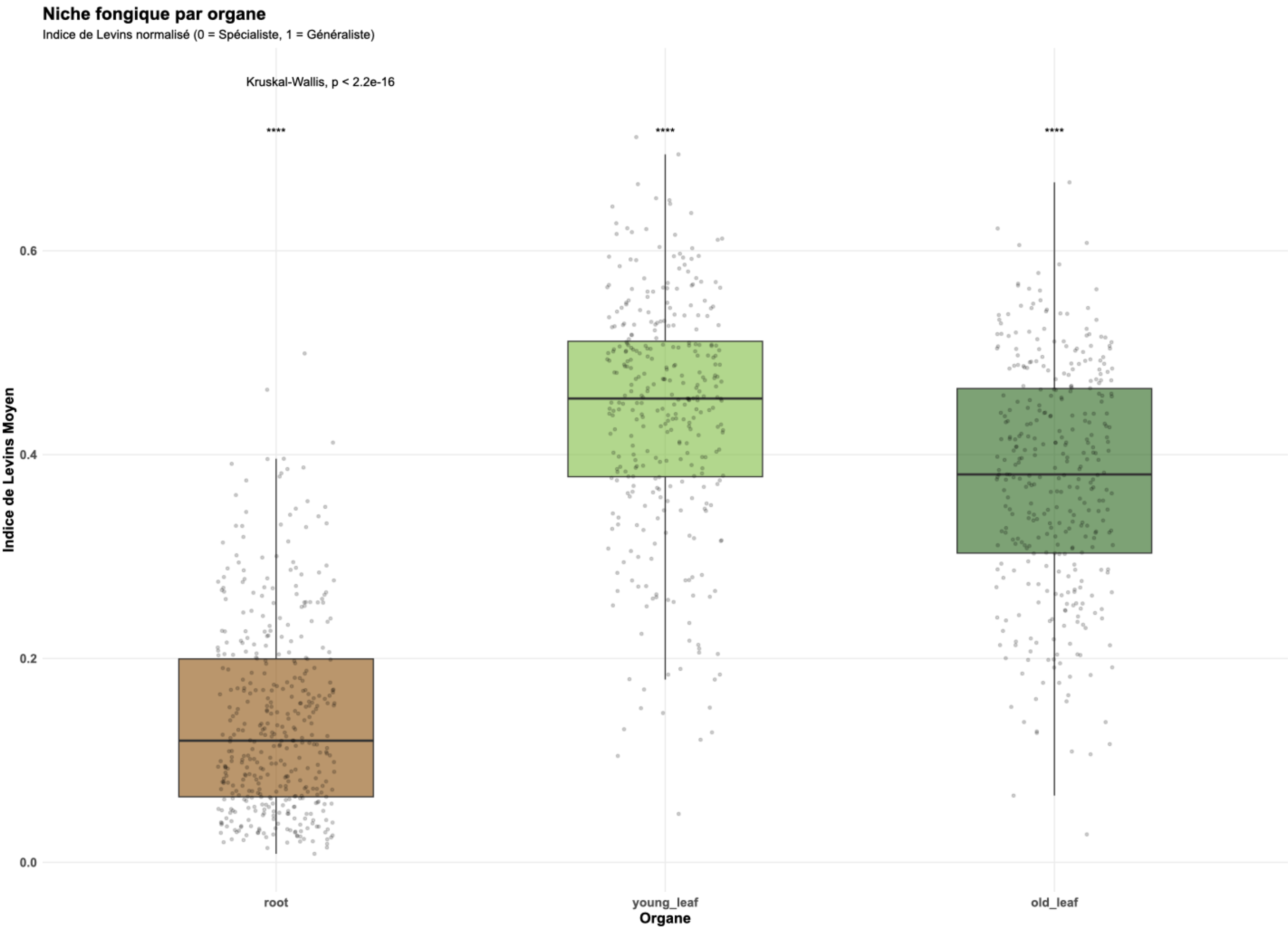


On retrouve le même diagramme que précédemment ! (C'est le même)



Graphique très intéressant on voit bien les familles dominantes (au seuil 1% et couleur Orange jaune) et les familles présentes partout mais en faible abondance (orange au seuil 0.1% et bleu au seuil 0.5%) à détailler et pourquoi pas ajouter un arbre pour embrancher les familles

Indice de Levins (B)



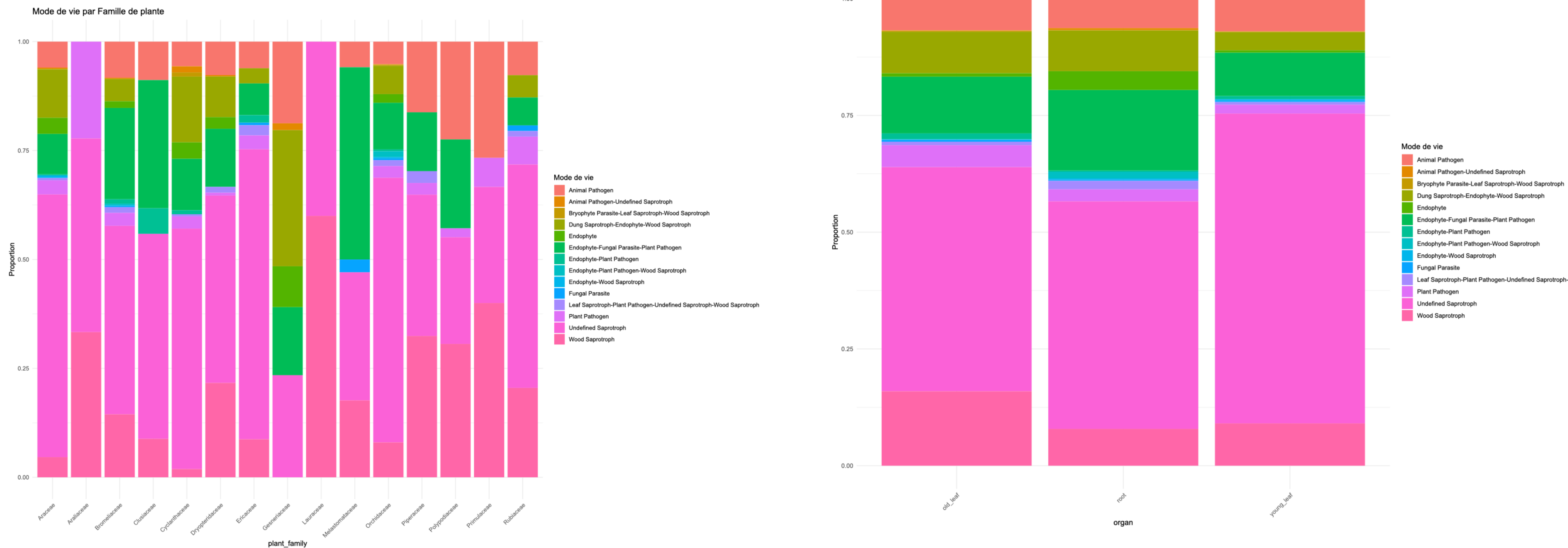
Summary

#	A tibble:	3	× 4		
	organ		Moyenne	SD	n
	<fct>		<dbl>	<dbl>	<int>
1	old_leaf		0.377	0.112	354
2	young_leaf		0.440	0.115	336
3	root		0.140	0.0946	417

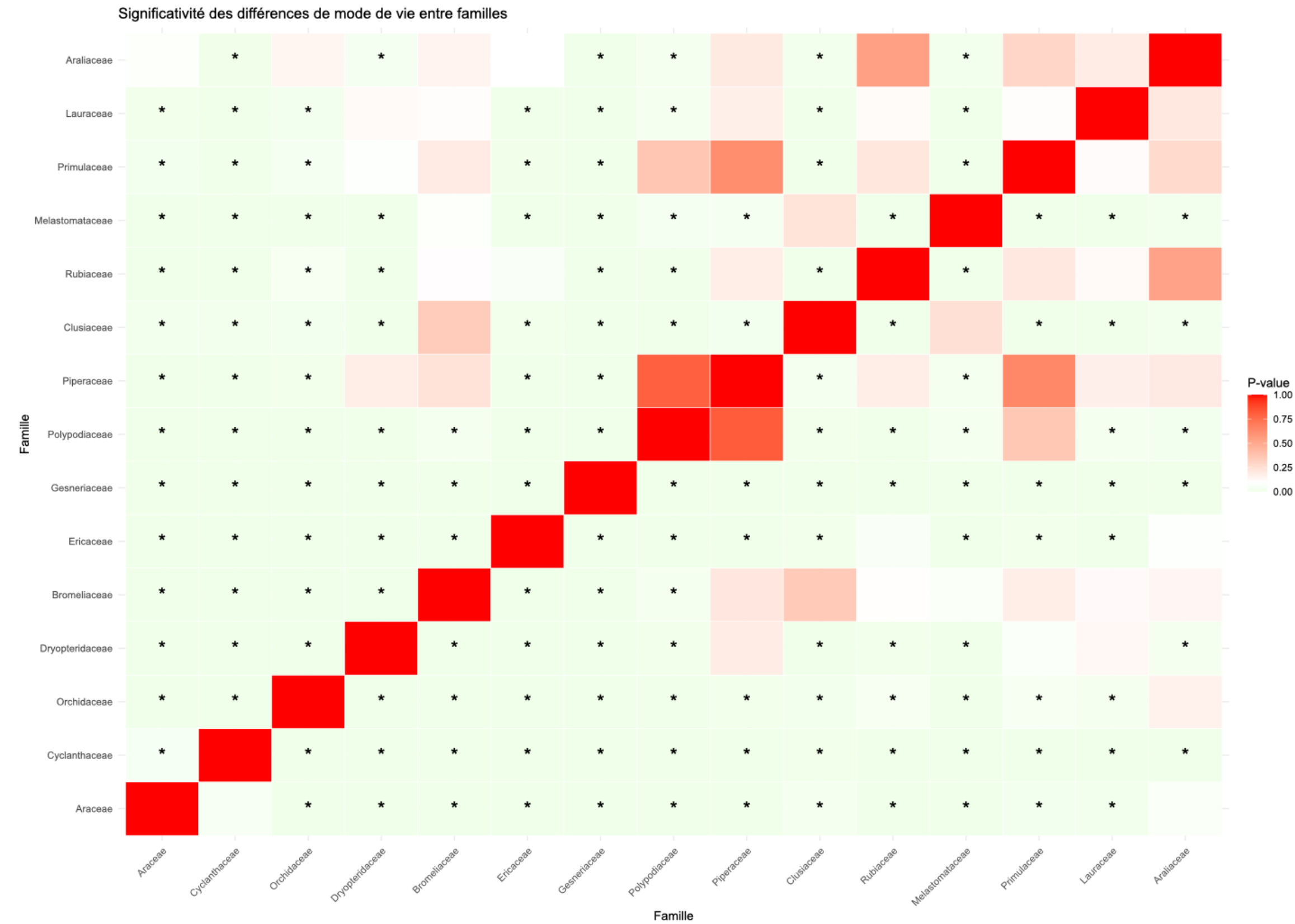
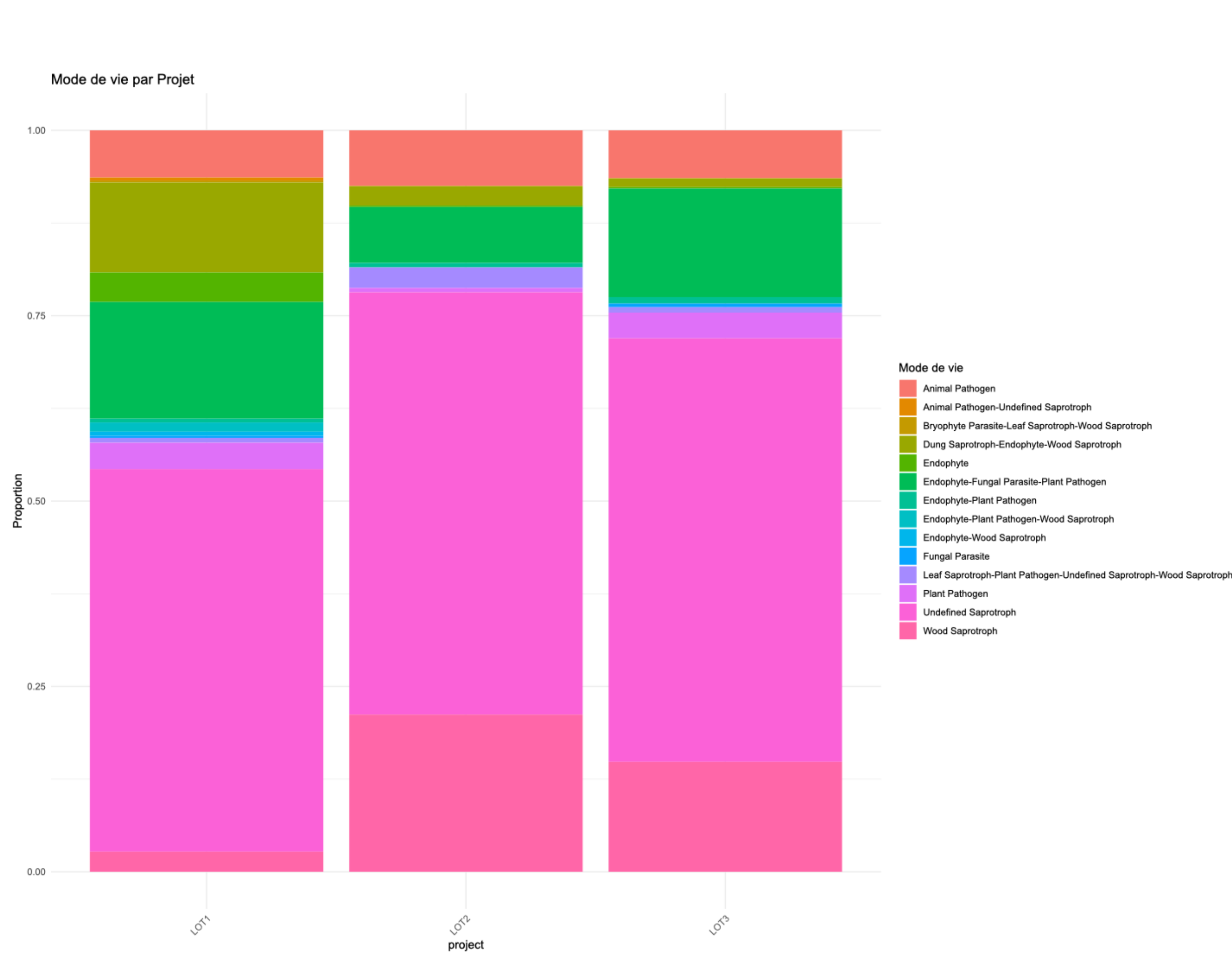
Graphique
interessant qui
montre que les tou
des racines sont
significativement
plus spécialistes
que ceux des
feuilles

Analyse des traits fonctionnels

Utilisation de Fungal Traits



Rien de surprenant ici, on retrouve toujours majoritairement des saprotrophes et des parasites de plantes... Pas sur que ce soit tres pertinent, surtout en sachant que seuls 225 tou sur 1107 ont une correspondance sur Fungal Traits



--- Test : Famille vs Mode de vie ---

Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

data: tbl

X-squared = 1043.7, df = NA, p-value = 0.0004998

Analyse canonique des corrélations (ACC)

```
r$> cca_mod <- cca(df_fonctions_mat ~ organ + project + plant_family, data = metadata_sub)

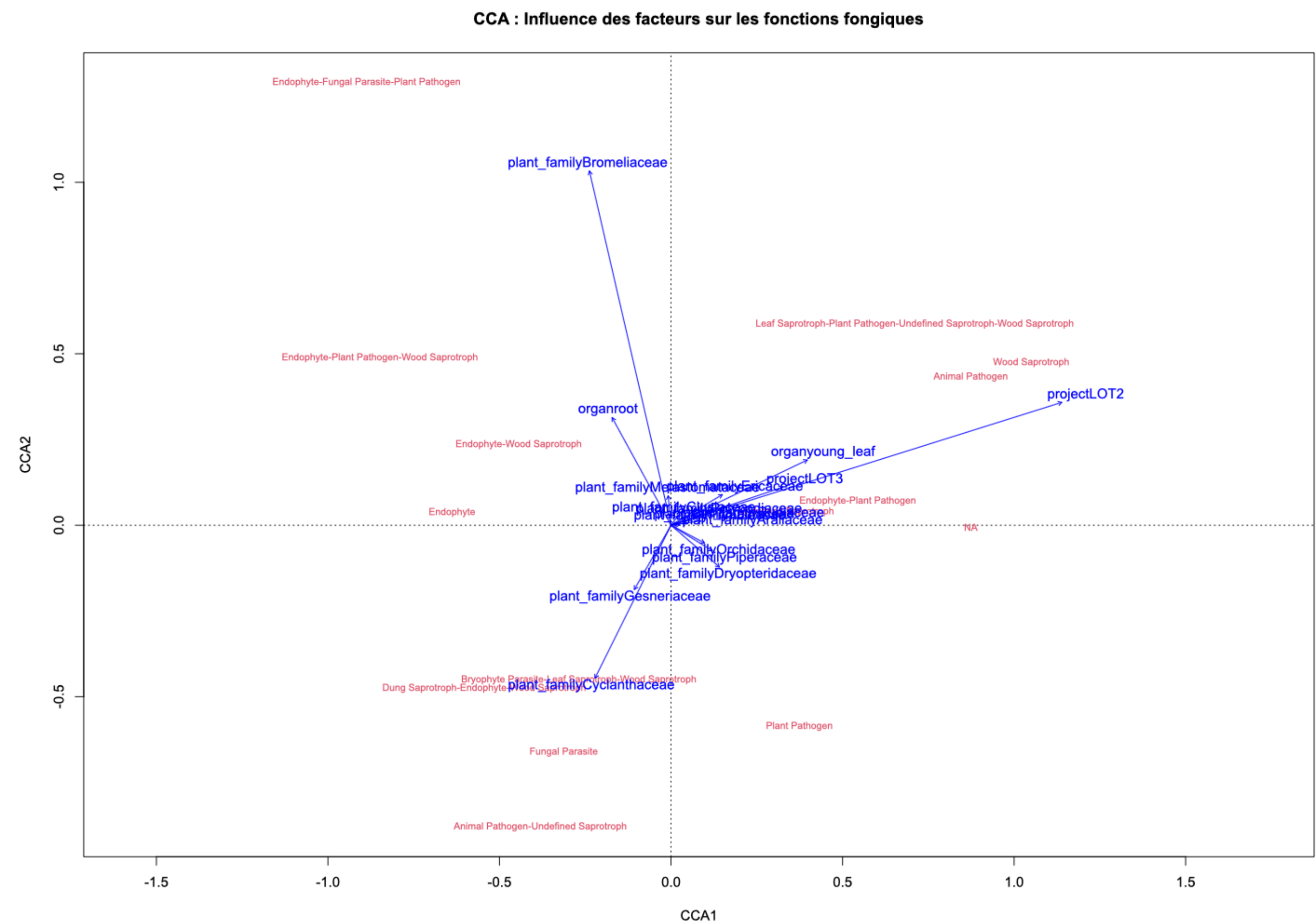
r$> anova(cca_mod, by = "term")
Permutation test for cca under reduced model
Terms added sequentially (first to last)
Permutation: free
Number of permutations: 999

Model: cca(formula = df_fonctions_mat ~ organ + project + plant_family, data = metadata_sub)
      Df ChiSquare      F Pr(>F)
organ    2    0.2813 23.1596 0.001 ***
project  2    0.4907 40.4026 0.001 ***
plant_family 14    0.4861  5.7186 0.220
Residual 962    5.8415

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

r$> print(scores_cca)
```

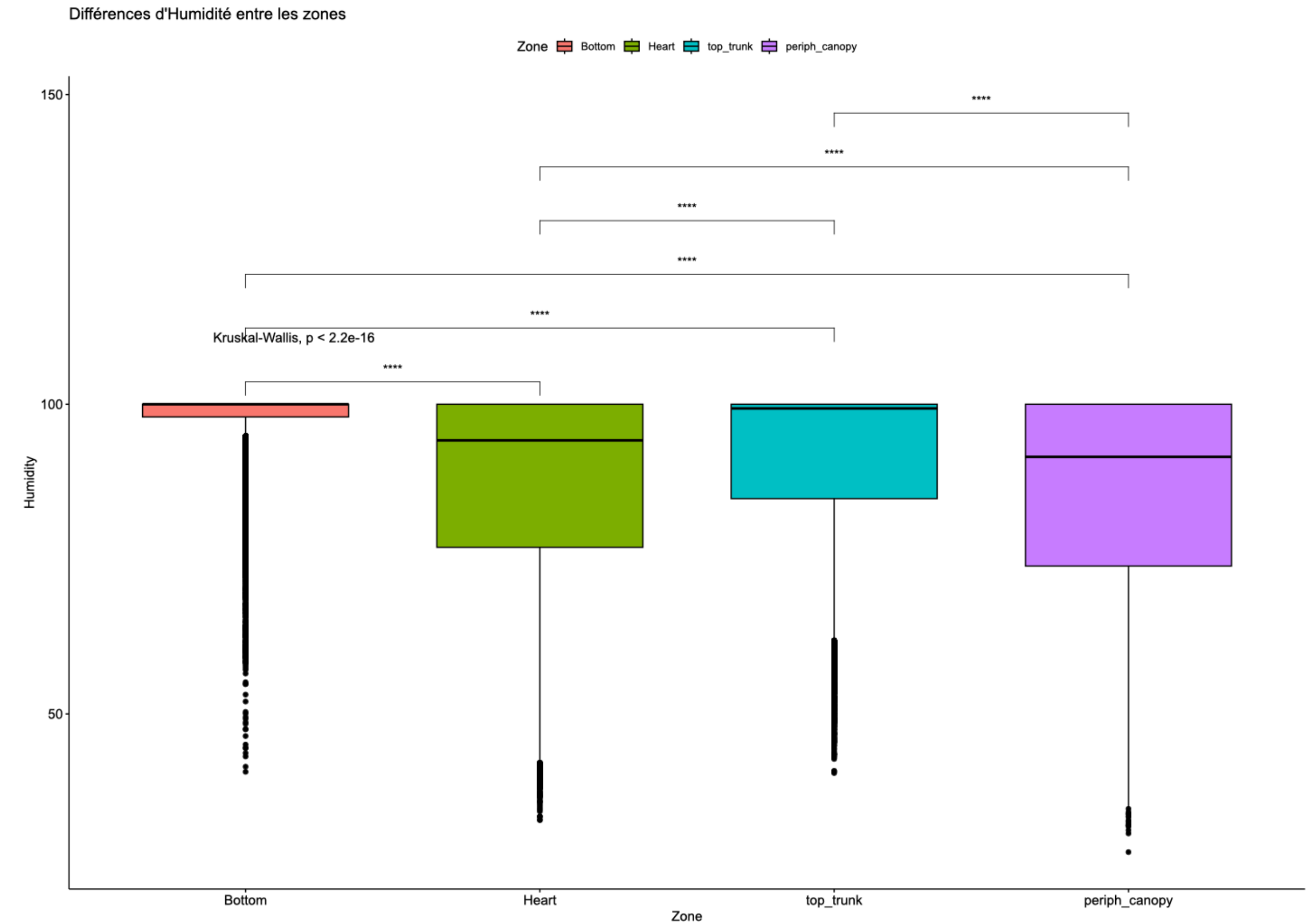
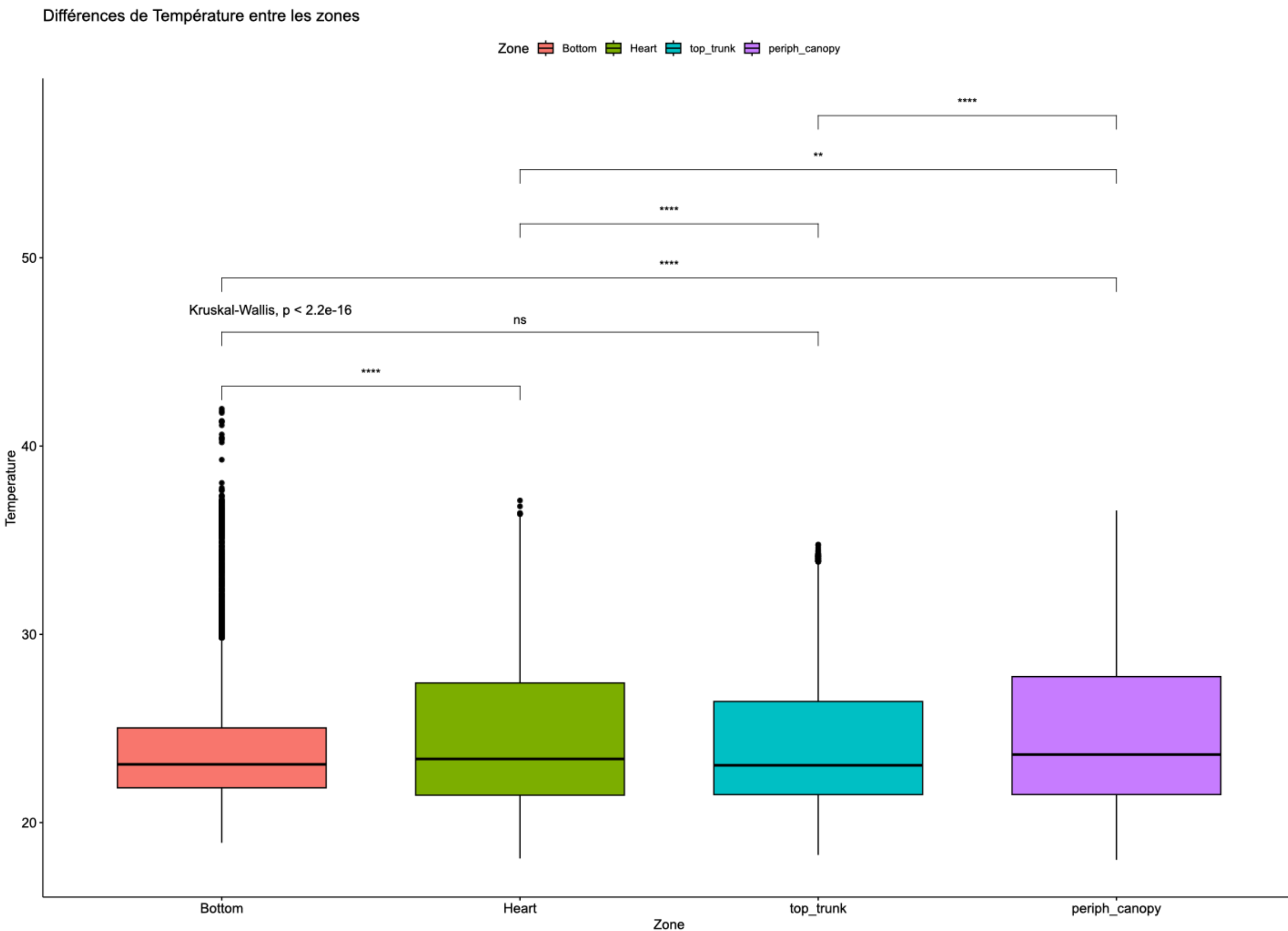
	CCA1	CCA2
organroot	-0.12514726	0.2287148313
organyoung_leaf	0.28950032	0.1386545380
projectLOT2	0.82996800	0.2610134551
projectLOT3	0.23738973	0.0809383144
plant_familyAraliaceae	0.01846631	0.0009955989
plant_familyBromeliaceae	-0.17332678	0.7528486165
plant_familyClusiaceae	0.01520892	0.0207114985
plant_familyCyclanthaceae	-0.16151660	-0.3250022910
plant_familyDryopteridaceae	0.10275298	-0.0885220735
plant_familyEricaceae	0.10897247	0.0649908762
plant_familyGesneriaceae	-0.07796521	-0.1365652442
plant_familyLauraceae	0.03249132	0.0063906642
plant_familyMelastomataceae	-0.00611200	0.0626997446
plant_familyOrchidaceae	0.07098788	-0.0376581651
plant_familyPiperaceae	0.08773575	-0.0542693011
plant_familyPolypodiaceae	0.05382934	0.0178163151
plant_familyPrimulaceae	0.01515549	0.0035934823
plant_familyRubiaceae	0.06352557	0.0090436077



Assez intéressant, on voit que les broméliacées sont plus propices à êtres parasitées, de même que les Cyctanthaceae. Le LOT2 est plus propices pour la présence de pathogènes d’animaux, peut-être arbre accueillant plus de biodiversité d’animaux ?

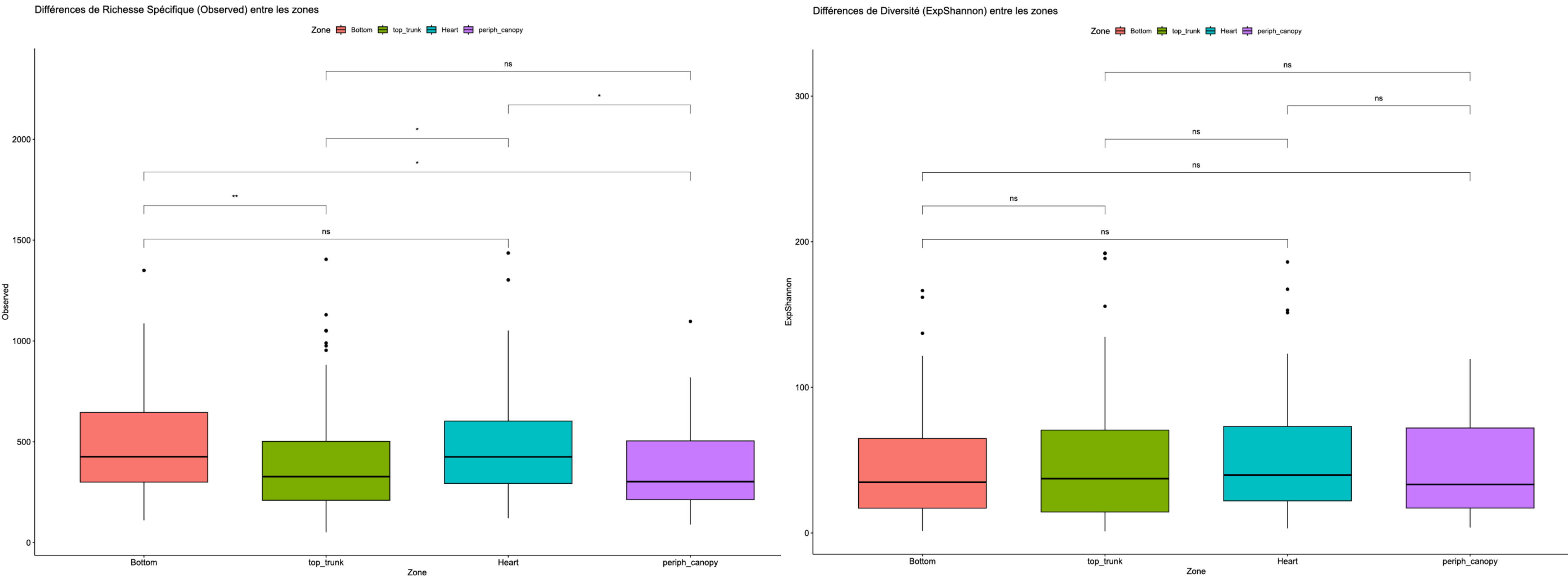
Croisement phyloseq x data env

Avec les données du LOT 1 - diff. env. / zones



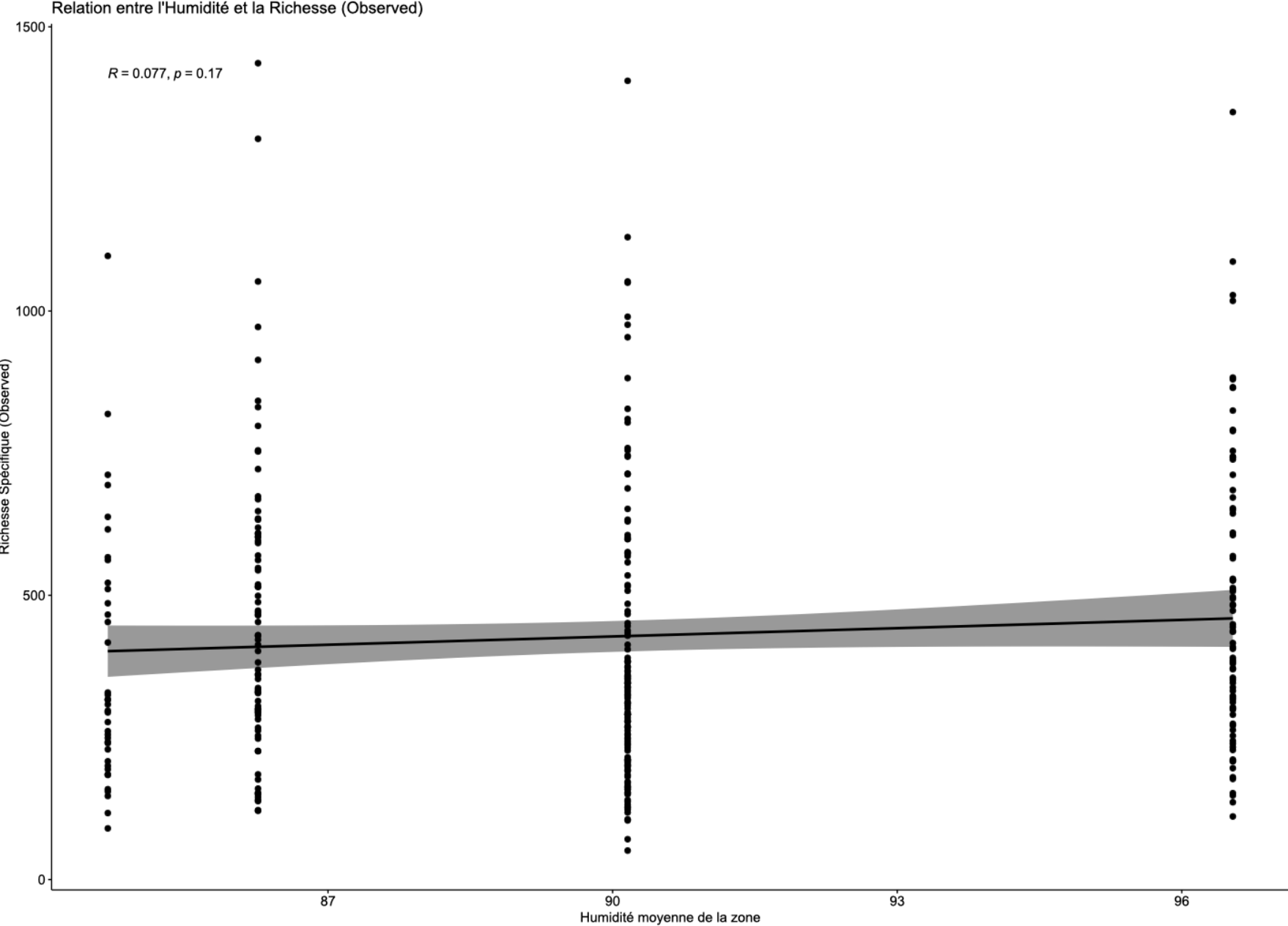
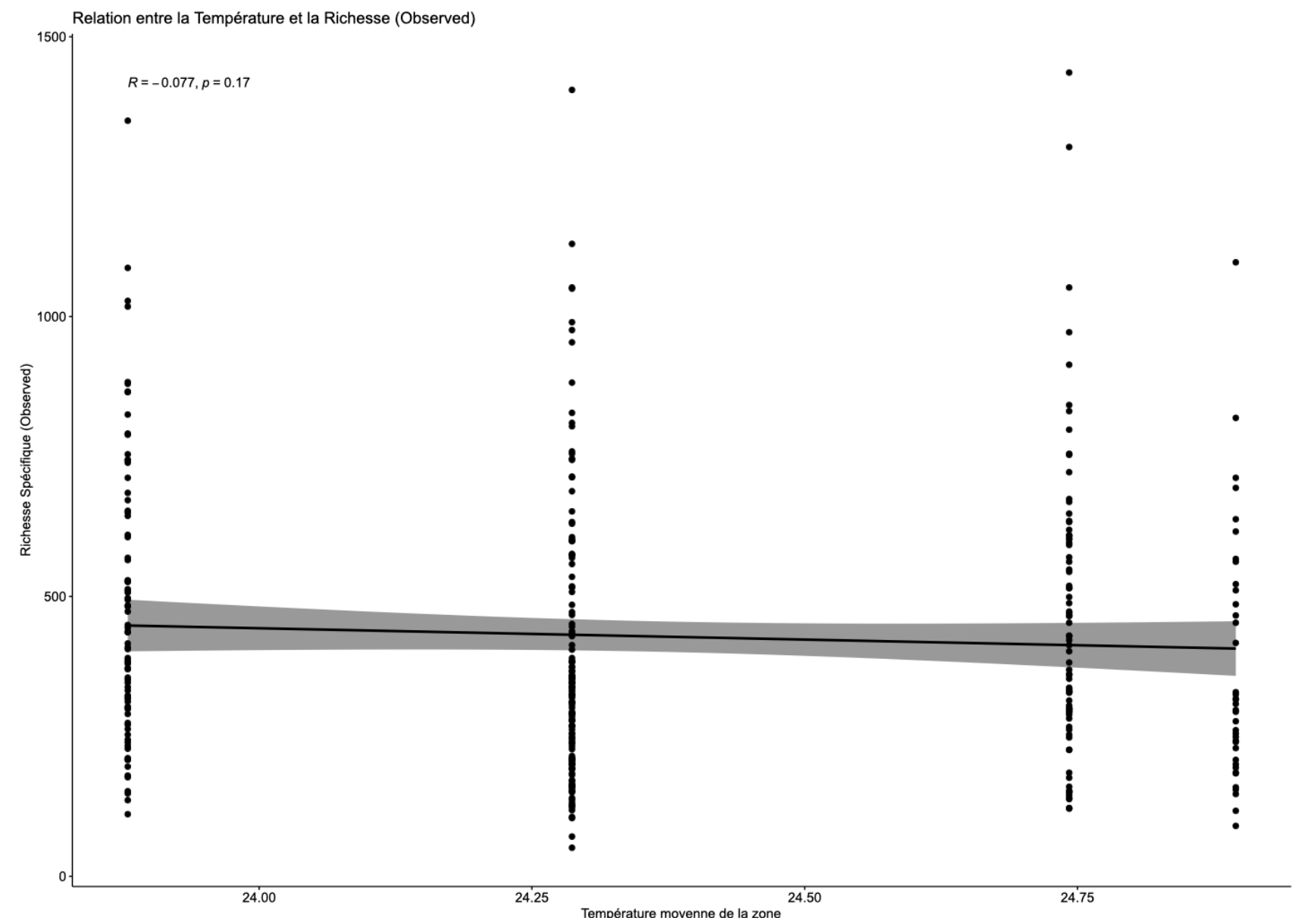
Globalement les paramètres environnementaux ne sont pas les mêmes entre les zones d'échantillonnage au sein du LOT1, enthousiasmant pour la suite de l'étude

Differences de q=0 et q=1 par zones

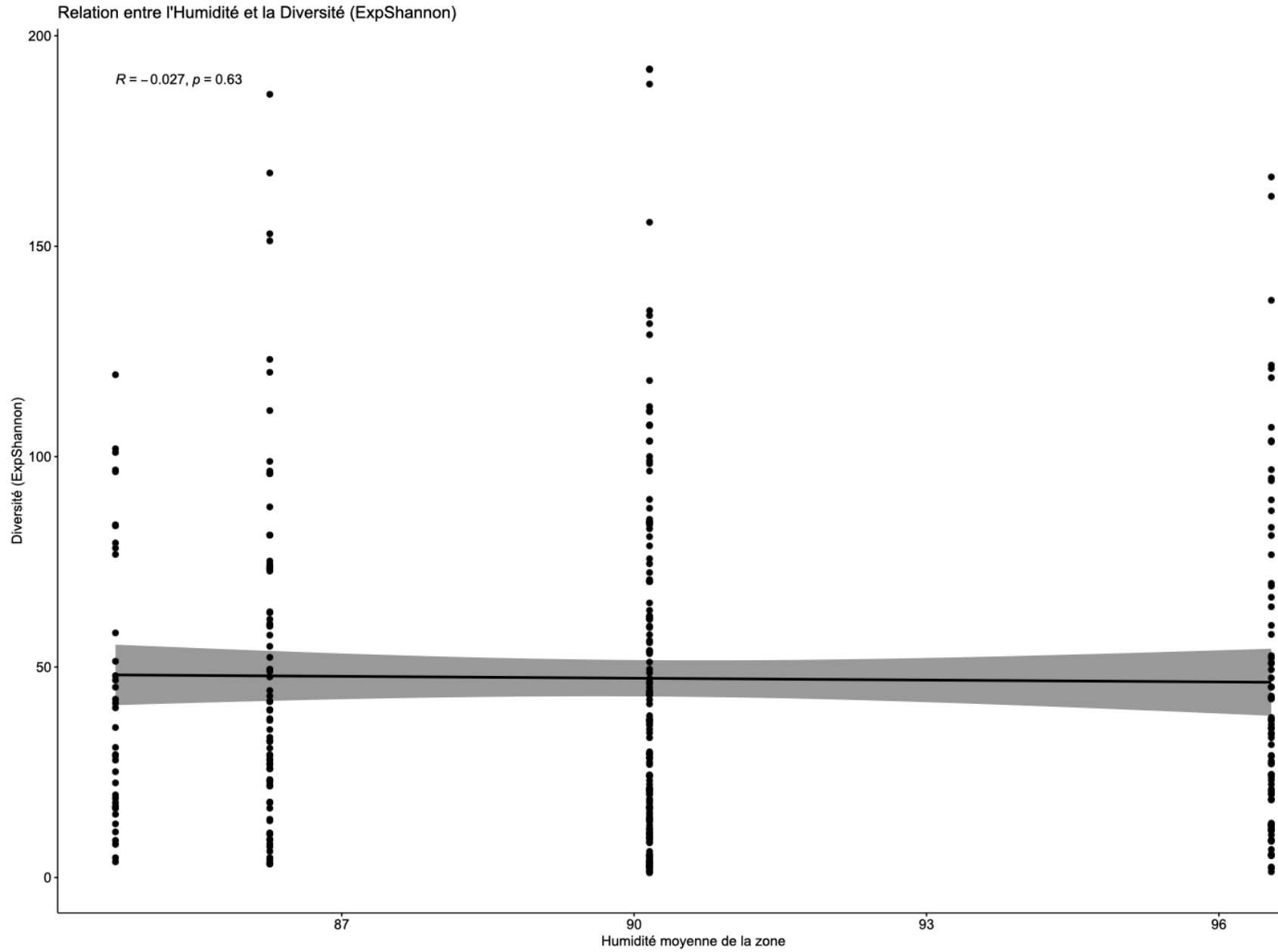
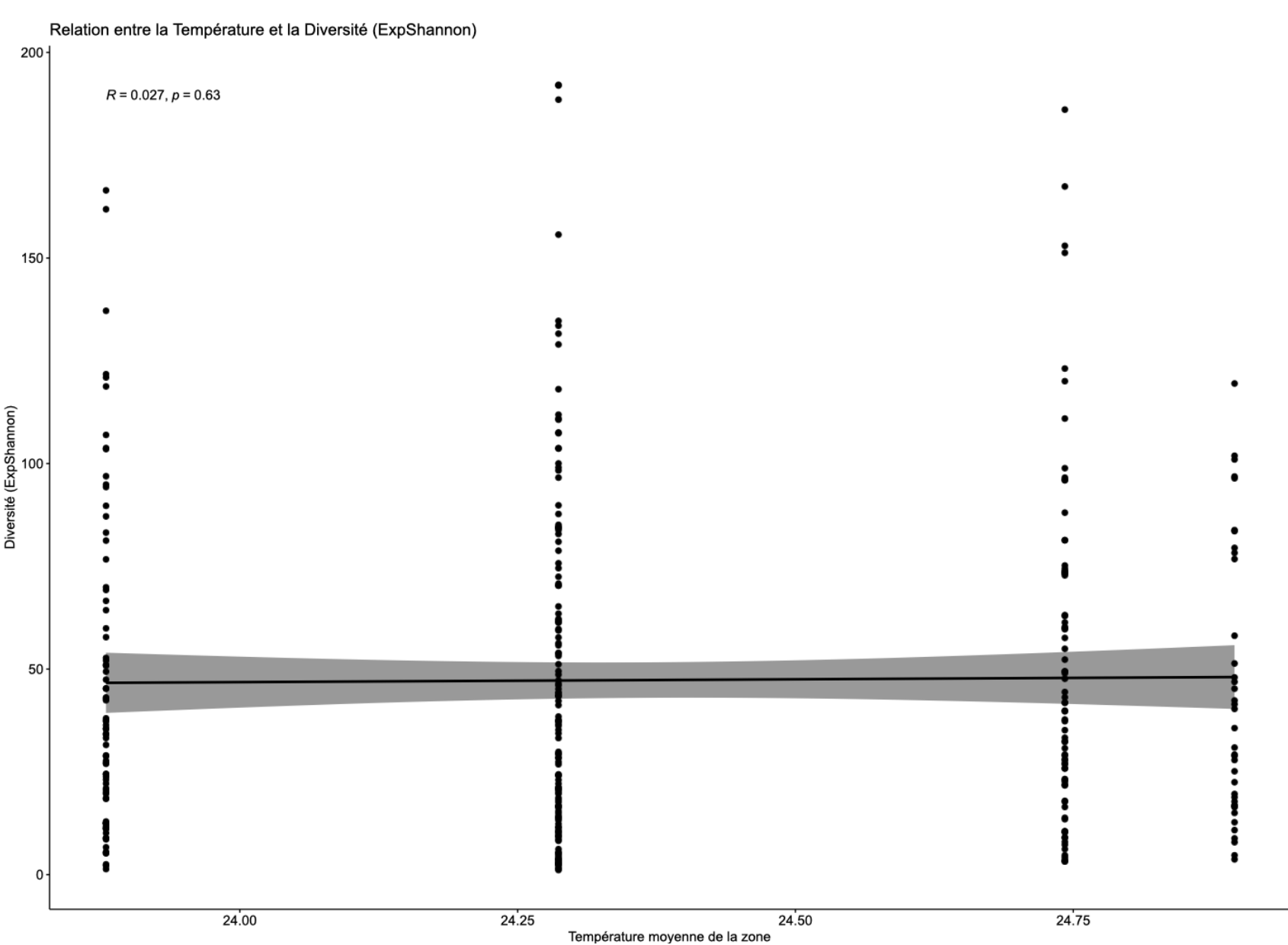


On observe des différences significatives (faibles) de rs entre la plupart des zones. Pas de différences de diversité

Correlations q avec env



Corrélation non
significative mais négative
entre température et
richesse et positive entre
humidité et rs. Pas
exploitable pour la
diversité. Maintenant il
faut tester avec un
modele ajusté



Test avec glm (négative binomial)

```
r$> glm_humi_temp_obs <- glm.nb(Observed ~ Mean_Humi * Mean_Temp, data = df_richness)
r$> summary(glm_humi_temp_obs)

Call:
glm.nb(formula = Observed ~ Mean_Humi * Mean_Temp, data = df_richness,
        init.theta = 3.323696435, link = log)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    262.70346   139.57495    1.882   0.0598 .
Mean_Humi      -4.03583     1.97777   -2.041   0.0413 *
Mean_Temp     -12.01146     6.24851   -1.922   0.0546 .
Mean_Humi:Mean_Temp  0.18215     0.08715    2.090   0.0366 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(3.3237) family taken to be 1)

    Null deviance: 357.58  on 329  degrees of freedom
Residual deviance: 346.14  on 326  degrees of freedom
AIC: 4477.3

Number of Fisher Scoring iterations: 1

              Theta:  3.324
            Std. Err.:  0.249

2 x log-likelihood:  -4467.302
```

```
r$> simulationOutput_humi_temp_obs <- simulateResiduals(fittedModel = glm_humi_temp_obs, plot = TRUE)
testUniformity(simulationOutput_humi_temp_obs)
testDispersion(simulationOutput_humi_temp_obs)
testZeroInflation(simulationOutput_humi_temp_obs)

Asymptotic one-sample Kolmogorov-Smirnov test

data:  simulationOutput$scaledResiduals
D = 0.054909, p-value = 0.2728
alternative hypothesis: two-sided

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

data:  simulationOutput
dispersion = 1.0844, p-value = 0.352
alternative hypothesis: two.sided

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

data:  simulationOutput
ratioObsSim = NaN, p-value = 1
alternative hypothesis: two.sided
```

```
r$> glm_humi_temp_exp <- glm.nb(ExpShannon ~ Mean_Humi * Mean_Temp, data = df_richness)
summary(glm_humi_temp_exp)
There were 50 or more warnings (use warnings() to see the first 50)

Call:
glm.nb(formula = ExpShannon ~ Mean_Humi * Mean_Temp, data = df_richness,
        init.theta = 1.352430258, link = log)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    179.6177   221.1007    0.812   0.417
Mean_Humi      -2.4297     3.1329   -0.776   0.438
Mean_Temp     -7.7928     9.8981   -0.787   0.431
Mean_Humi:Mean_Temp  0.1062     0.1380    0.769   0.442

(Dispersion parameter for Negative Binomial(1.3524) family taken to be 1)

    Null deviance: 366.77  on 329  degrees of freedom
Residual deviance: 366.04  on 326  degrees of freedom
AIC: 3206.6

Number of Fisher Scoring iterations: 1

              Theta:  1.352
            Std. Err.:  0.100

2 x log-likelihood:  -3196.553
```

```
r$> simulationOutput_humi_temp_exp <- simulateResiduals(fittedModel = glm_humi_temp_exp, plot = TRUE)
testUniformity(simulationOutput_humi_temp_exp)
testDispersion(simulationOutput_humi_temp_exp)
testZeroInflation(simulationOutput_humi_temp_exp)

Asymptotic one-sample Kolmogorov-Smirnov test

data:  simulationOutput$scaledResiduals
D = 0.033697, p-value = 0.8478
alternative hypothesis: two-sided

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

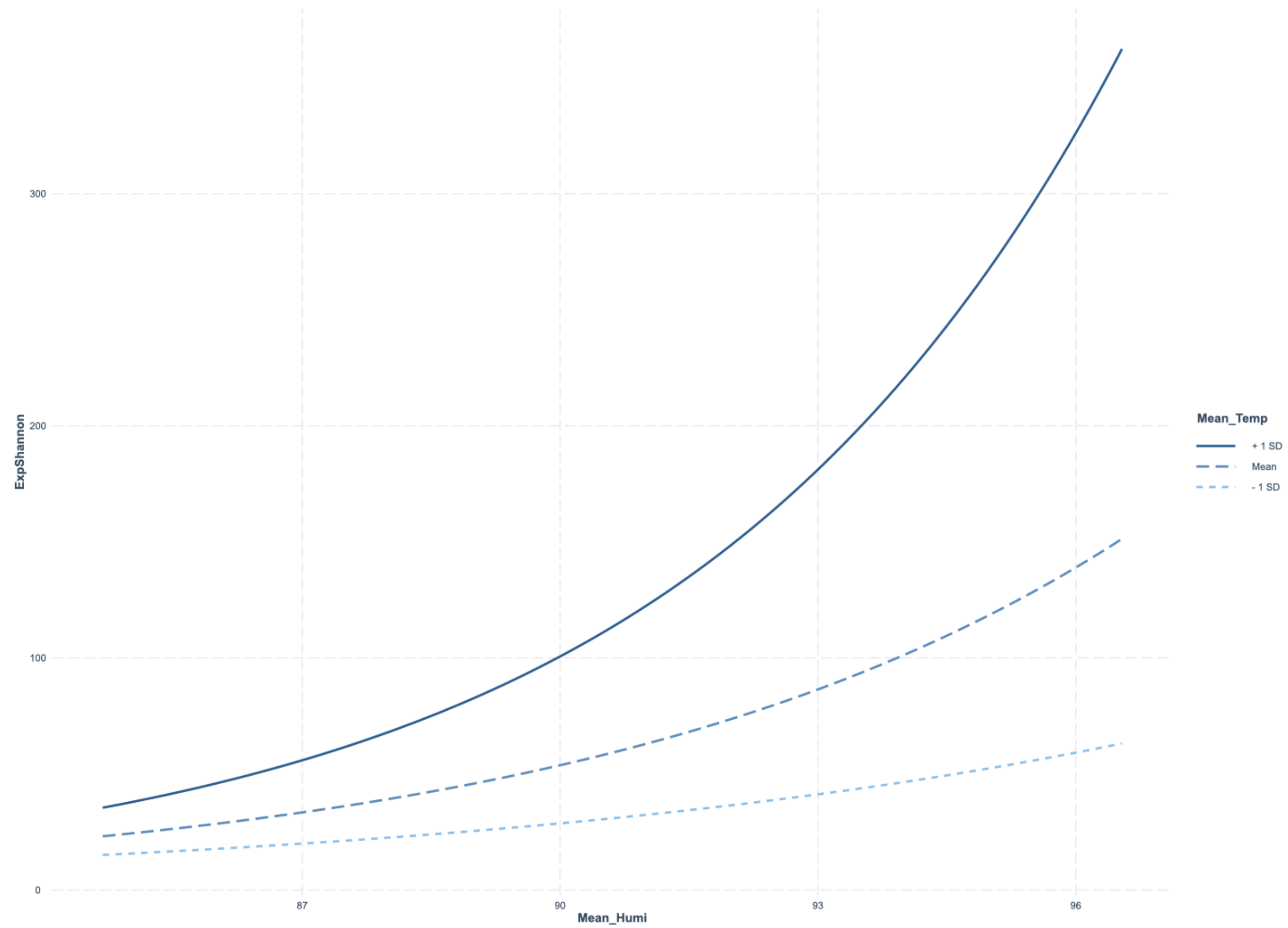
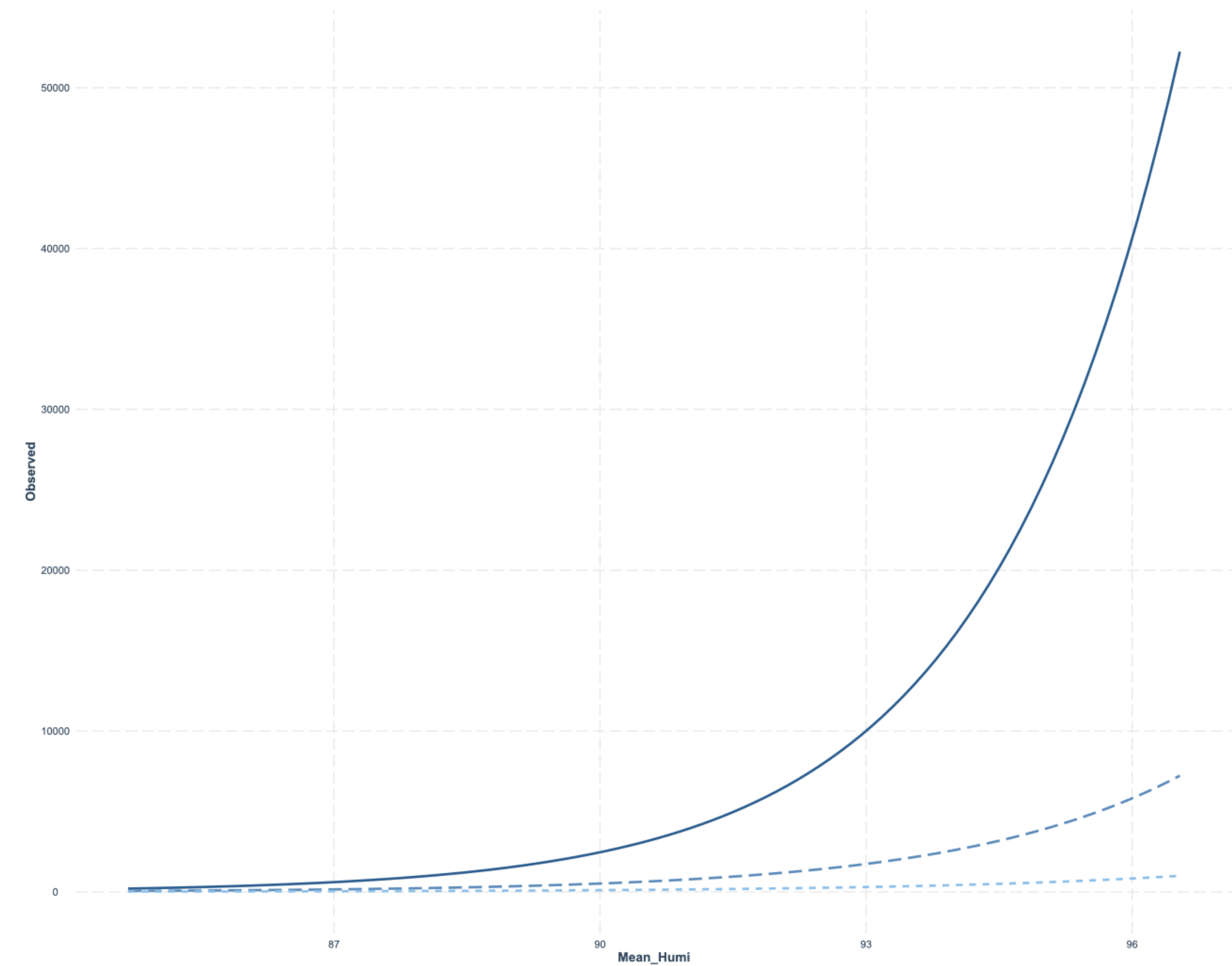
data:  simulationOutput
dispersion = 0.92064, p-value = 0.64
alternative hypothesis: two.sided

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model

data:  simulationOutput
ratioObsSim = 0, p-value = 0.16
alternative hypothesis: two.sided
```

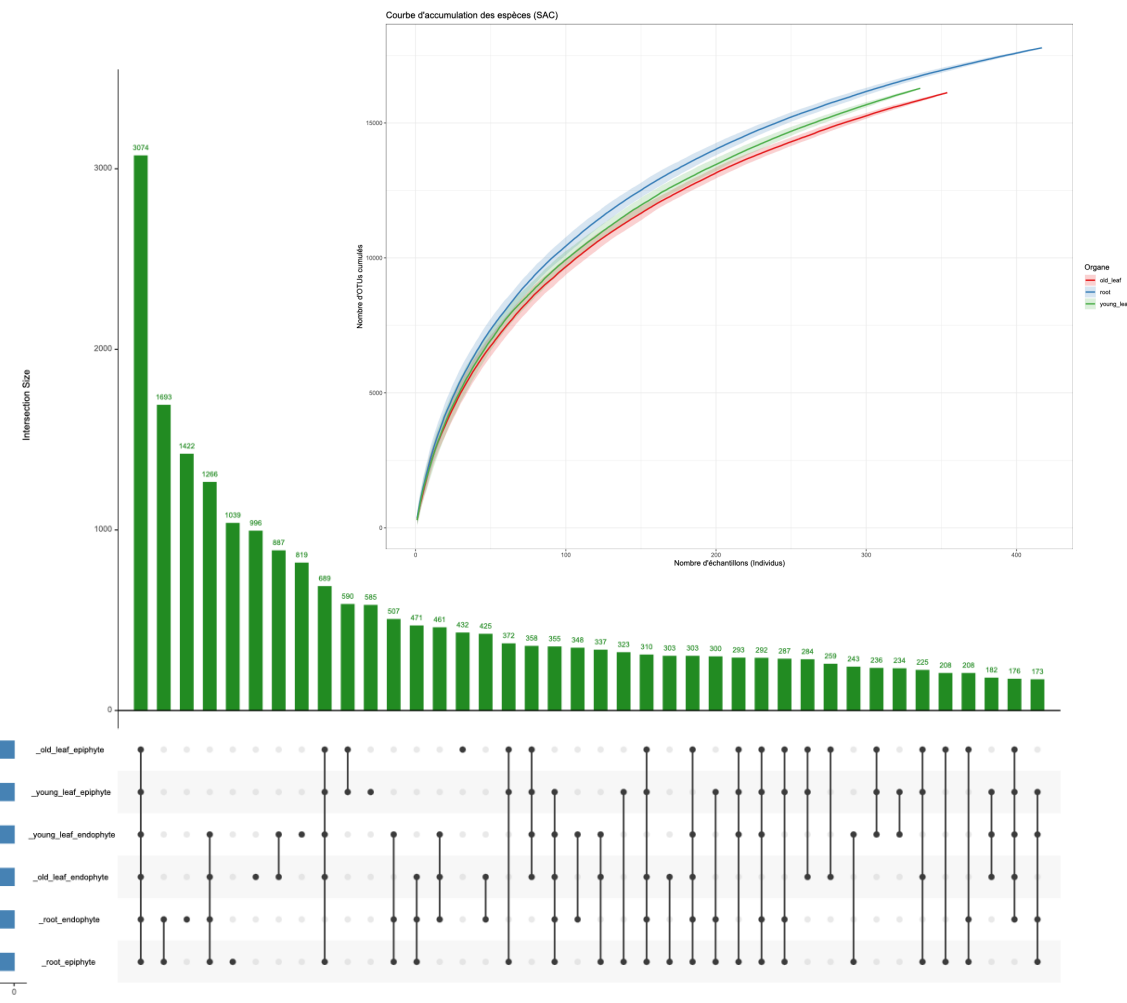
En bref: modele exploitable et significatif pour rs. L’humidité et l’interaction humidité et température à effectivement un effet positif sur la rs des communautés fongiques (ce qui paraît logique)

Graphiques d'interactions

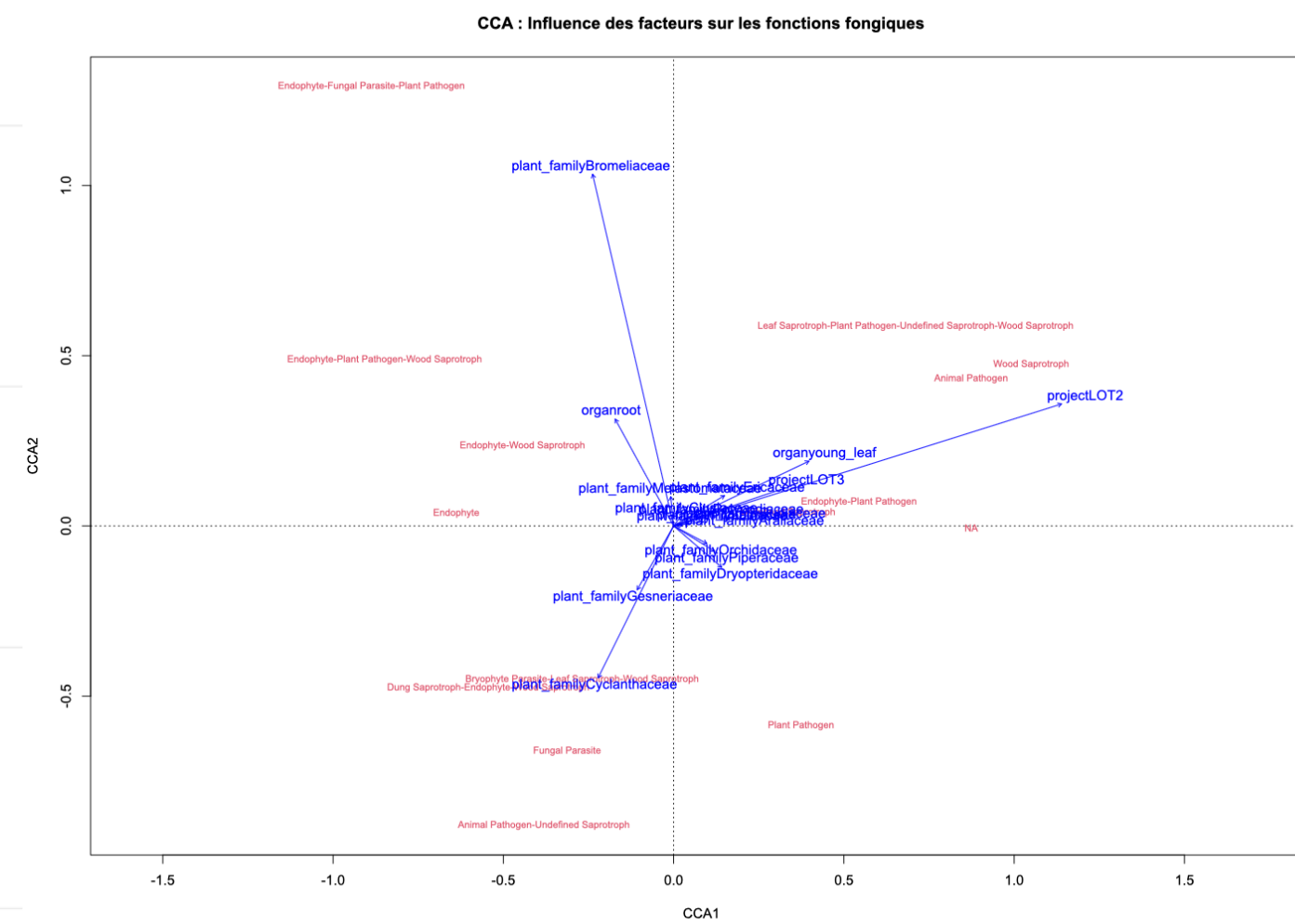
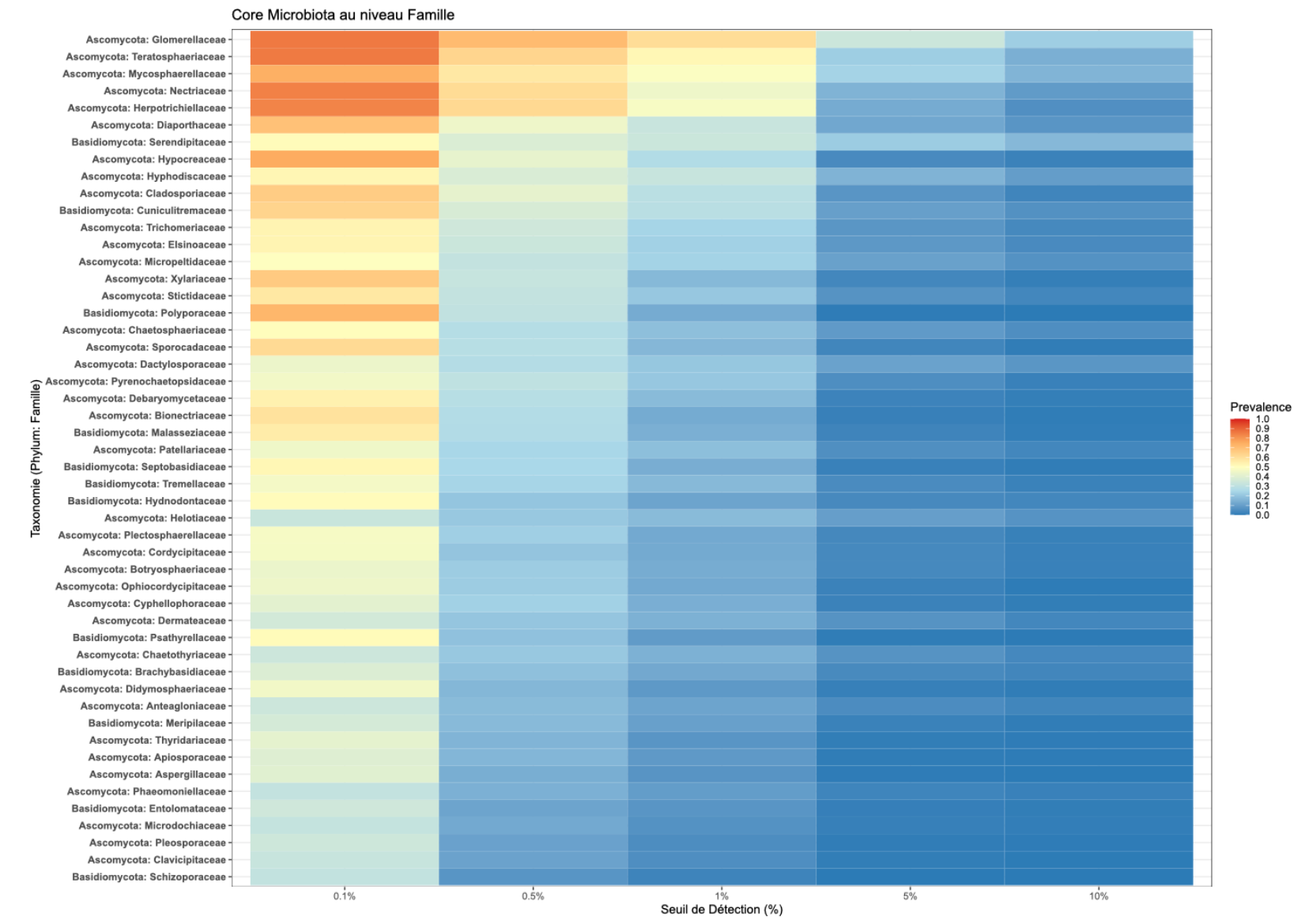
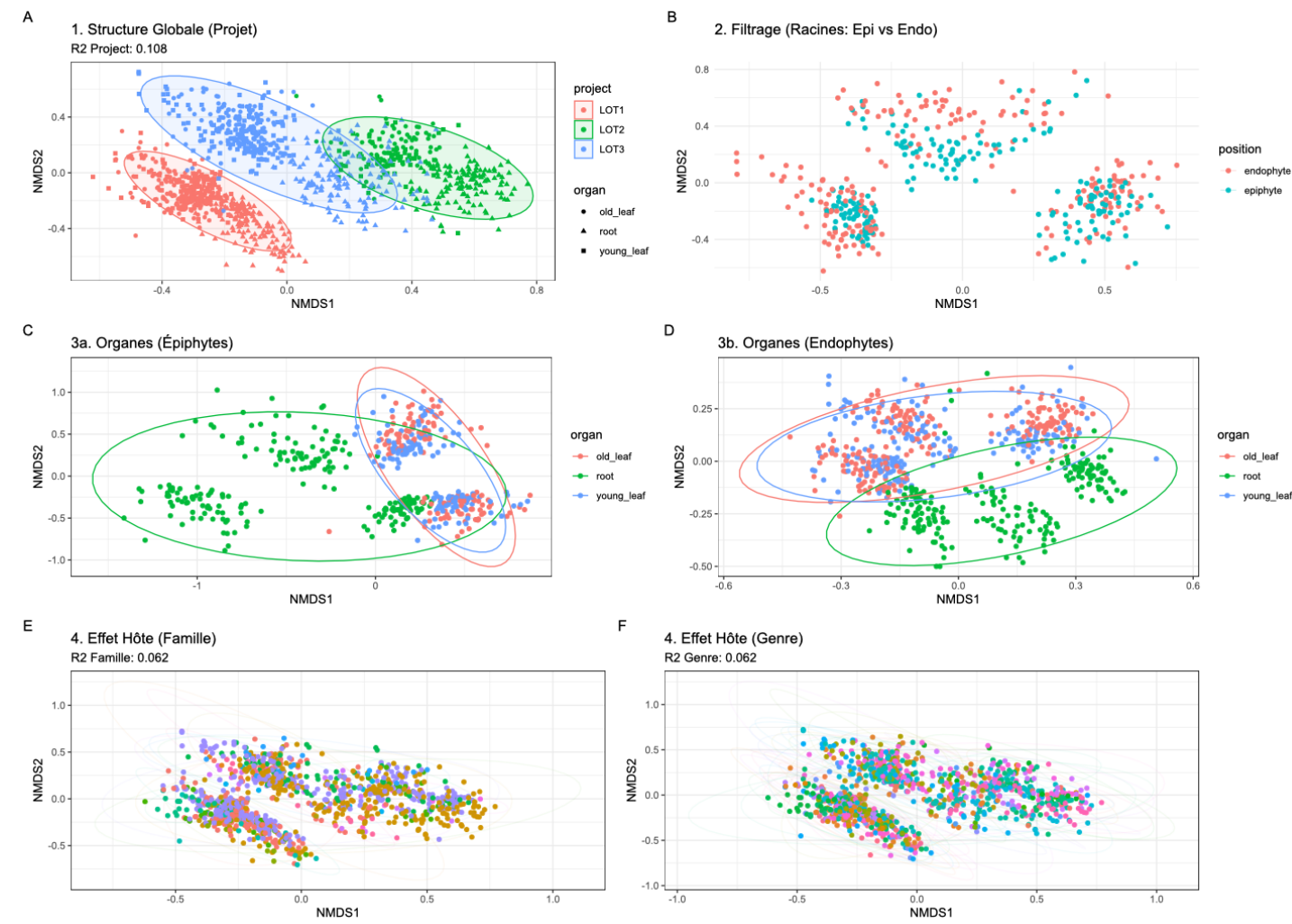
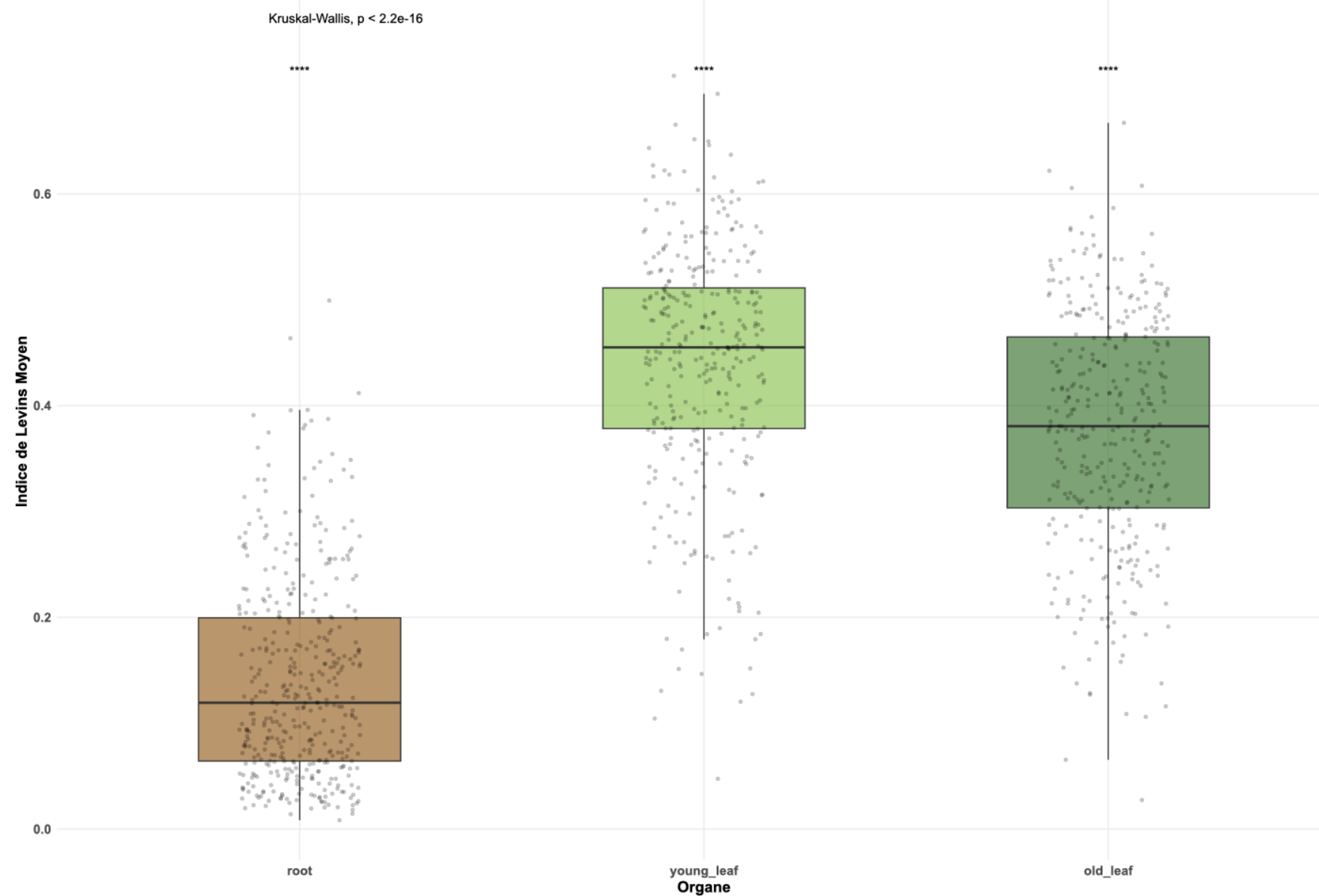


Une autre façon de le montrer, plus frappant mais moins rigoureux

Résumé des analyses pertinentes



Niche fongique par organe
Indice de Levins normalisé (0 = Spécialiste, 1 = Généraliste)



```
r$> glm_humi_temp_obs <- glm.nb(Observed ~ Mean_Humi * Mean_Temp, data = df_richness)

r$> summary(glm_humi_temp_obs)

Call:
glm.nb(formula = Observed ~ Mean_Humi * Mean_Temp, data = df_richness,
init.theta = 3.323696435, link = log)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    262.70346   139.57495    1.882   0.0598 .
Mean_Humi      -4.03583    1.97777   -2.041   0.0413 *
Mean_Temp     -12.01146    6.24851   -1.922   0.0546 .
Mean_Humi:Mean_Temp  0.18215    0.08715    2.090   0.0366 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(3.3237) family taken to be 1)

Null deviance: 357.58  on 329  degrees of freedom
Residual deviance: 346.14  on 326  degrees of freedom
AIC: 4477.3

Number of Fisher Scoring iterations: 1

                Theta: 3.324
                Std. Err.: 0.249

2 x log-likelihood: -4467.302
```

Résumé des analyses pertinentes

- La rs et div fluctue en fct des modalités
- L'échantillonnage est suffisant pour résultats robustes
- La majorité des otu sont partagés entre organes mais pas entre projets
- L'effet projet est très fort, organe et position faible et famille plante ns
- Les racines sont spécialistes
- Les modalités sélectionnent des traits de vies différents
- L'humidité explique la richesse spécifique fongiques dans le LOT1 (relation positive)

Pour poursuivre :

- Analyser données env pour tous les LOTs si possible
- Développer la partie core microbiota

Analyses biblio pistes

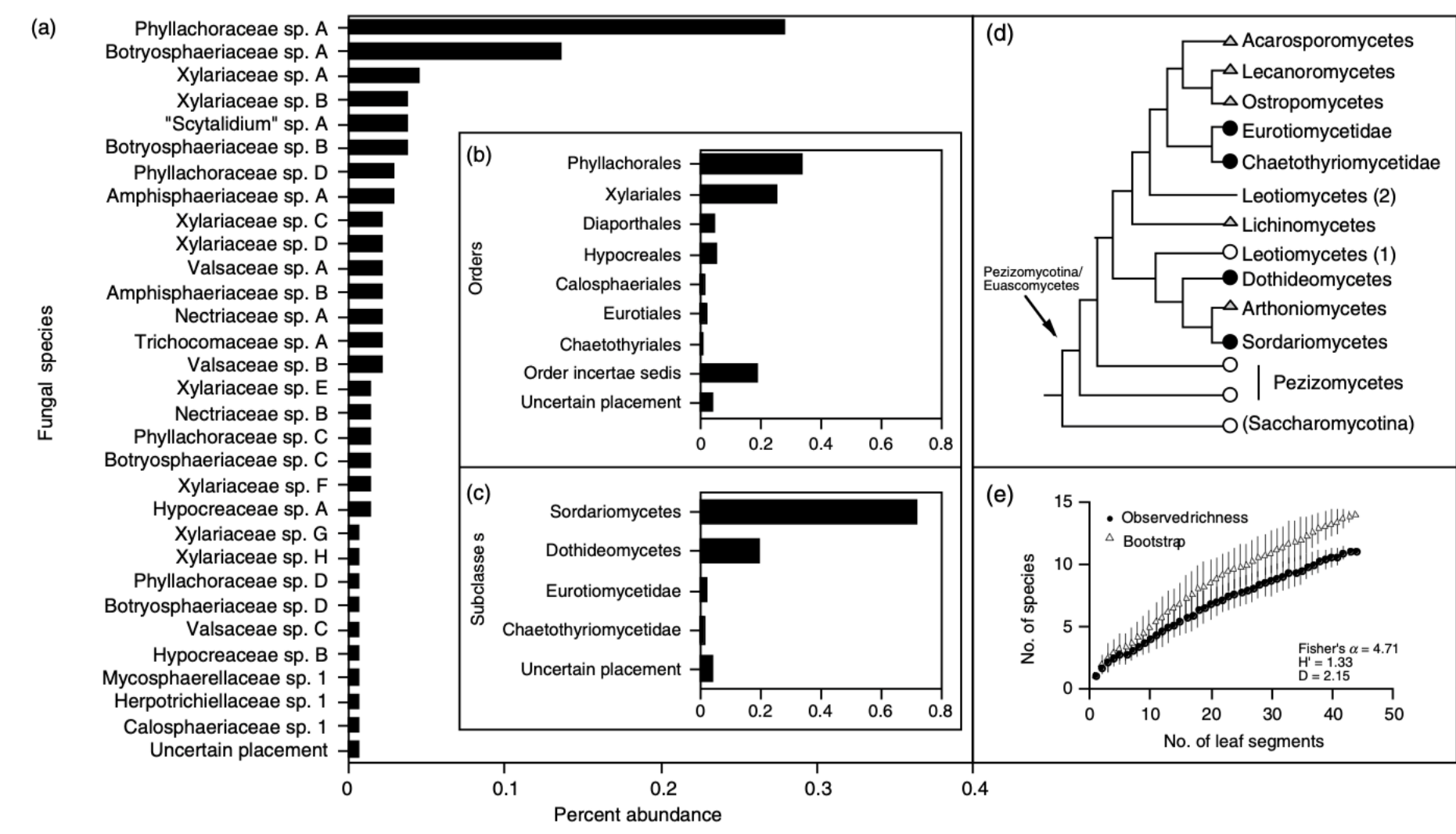


Figure 15.1 Taxonomic diversity and relative abundance of fungal endophytes inhabiting healthy leaves of tropical trees at Barro Colorado Island, Panama (BCI). (a) Relative abundance and taxonomic placement of the 31 most common species isolated from three mature leaves of *Laetia thamnia* (Flacourtiaceae), *Trichilia tuberculata* (Meliaceae), and *Gustavia superba* (Lecythidaceae). Taxonomic placements are based on BLAST matches in the NCBI GenBank database for sequence data (ca. 600 base pairs) from the nuclear ribosomal ITS regions, including the 5.8S gene ($N = 127$ isolates), coupled with phylogenetic analyses (Arnold and Lutzoni 2007). Most species are rare, and very few species are common. Panels (b) and (c) summarize these data at the ordinal (b) and subclass (c) levels, demonstrating the dominance of the Sordariomycetes (especially Phyllachorales and Xylariales). Panel (d) shows the current phylogenetic hypothesis for relationships of the Euascomycota (after Lutzoni *et al.* 2004), showing the phylogenetic breadth of endophytic isolates from only nine leaves at BCI. Black circles indicate lineages in which endophytes were found in the present study, white circles indicate lineages in which endophytes are known from other studies, and gray triangles indicate lineages in which the majority of (Arthoniomycetes) or all known species (e.g., Lichinomycetes) are lichenized, rather than free-living. With the exception of the clade of decomposer fungi known as Leotiomycetes (2), all major lineages that are not lichenized contain endophytic fungi. Panel (e) indicates the accumulation of endophyte species, defined as indicated in (a), as a function of the number of leaf segments sampled from leaves of *Laetia thamnia*, *Trichilia tuberculata*, and *Gustavia superba* ($N = 3$ leaves/species, 15 tissue segments/leaf). Even when singletons are excluded from the analysis, the accumulation of observed richness (black circles) and estimated richness based on bootstrap analyses (triangles) continues to rise, and diversity values (Fisher's α , Shannon index [H'], and Simpson's index [D]) remain high.

Arnold, A. E. (2008). Endophytic fungi: hidden components of tropical community ecology. *Tropical forest community ecology*, 178-188.

Visualisation condensée et complète

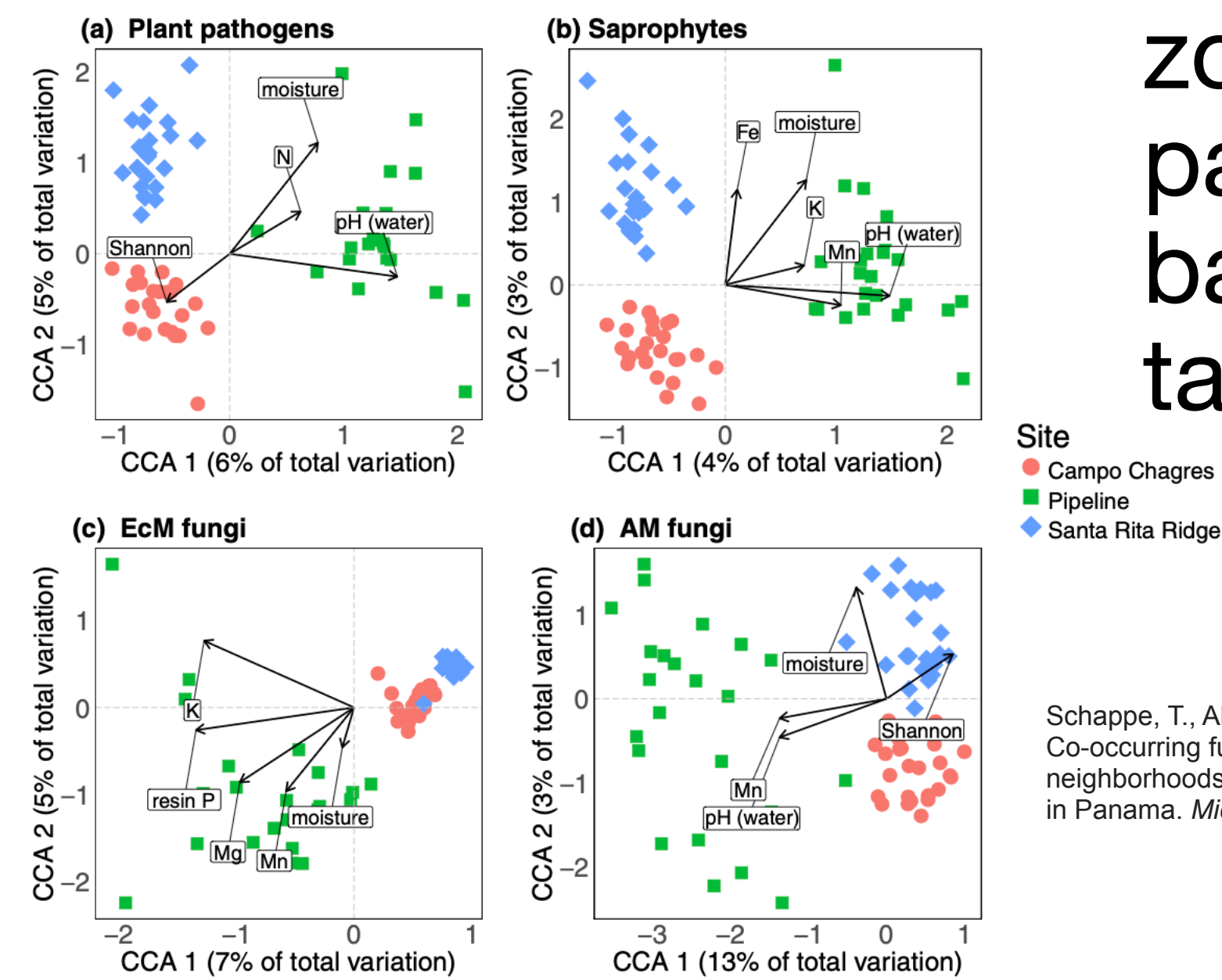
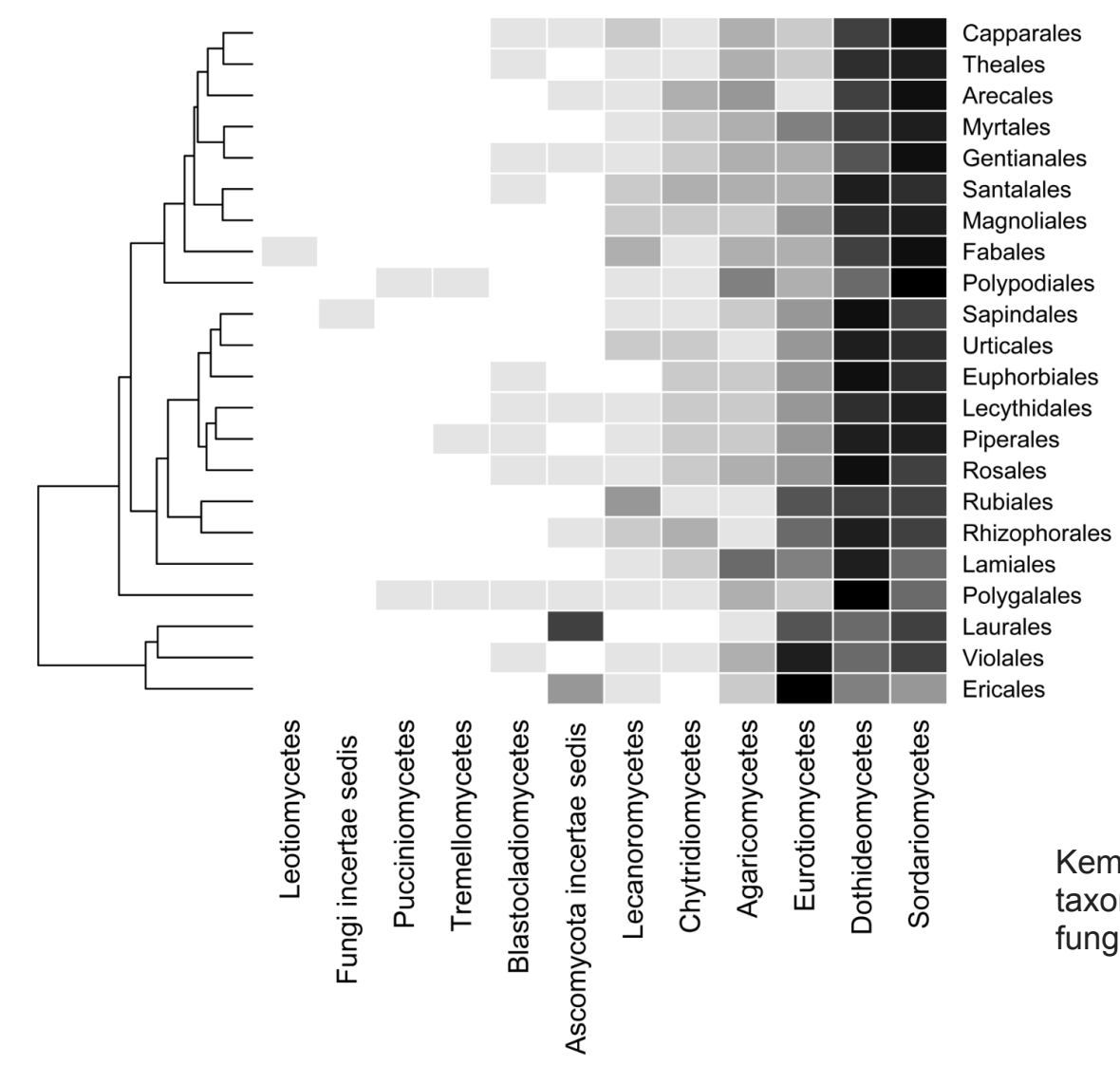


Fig. 3. Relative abundance (black = high abundance, white = low abundance) of abundant fungal classes (represented by >100 sequences) on leaves of host plants from different orders in a tropical forest in Panama. The dendrogram indicates the overall compositional similarity (complete linkage clustering of Bray–Curtis dissimilarity) of fungal communities on leaves from different plant host orders.



Kembel, S. W., & Mueller, R. C. (2014). Plant traits and taxonomy drive host associations in tropical phyllosphere fungal communities. *Botany*, 92(4), 303-311.

Analyse traits x zones après passage dans banque taxonomie

Schappe, T., Albornoz, F. E., Turner, B. L., & Jones, F. A. (2020). Co-occurring fungal functional groups respond differently to tree neighborhoods and soil properties across three tropical rainforests in Panama. *Microbial Ecology*, 79(3), 675-685.

Regrouper les otu pour faire apparaitre d'éventuels pattern